

A novel method for online real-time forecasting of crude oil price

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HIGHLIGHTS

- A novel method is proposed to predict crude oil price.
- A novel optimization method of variational mode decomposition parameters is proposed.
- A new method for quantifying data uncertainty is proposed.
- The proposed method has good prediction accuracy and robustness
- The proposed method can be used for both point prediction and interval prediction.

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ABSTRACT

Improving the accuracy of crude oil price forecasting is helpful for stabilizing financial markets and oil import and export trade. However, the extant models rarely focus on the online forecasting and uncertainty of crude oil price. Motivated by these, a new forecasting method is proposed. Firstly, the price sequence is decomposed to get the regular sub-sequences and noise sequence by the improved variational mode decomposition, whose parameters are optimized by particle swarm optimization based on average maximum envelope entropy. Secondly, the proposed hybrid model of point prediction is established by the characteristics of sub-sequences, and the interval prediction model is constructed by combining the point prediction model and Bootstrap sampling. Finally, the proposed point prediction model and interval prediction model are formed into a predictor to realize online real-time prediction. The empirical results show that, compared with other competing models, the proposed model for point prediction improves basically by over 10% on loss functions in the different frequency data, which verifies that it is superior to other models in accuracy and robustness. The coverage rate and fluctuation consistency are almost all more than 70% in the interval prediction results. In a word, the proposed method has high flexibility and accuracy, which can provide referring information for practitioners working with oil.

1. Introduction

Crude oil is a crucial strategic resource, and its price and consumption affect the global macroeconomic trend [1] and the industry development [2]. Improving the prediction performance of crude oil prices is an increasingly worldwide hot issue and has many valuable applications. On a national level, crude oil has ordinary commodity property, and it is related to the development of industrial sectors. Crude oil price fluctuation deeply affects energy resource reserve decisions and international trade policy-making [3]. For example, crude oil volatility can

affect the risk exposure of oil industrial consumers and producers [4], and the rise in oil prices will lead to the inflation in oil-importing countries [5,6]. Thus, improving crude oil price forecasting helps to safeguard national security and facilitate economic growth. From a financial stability perspective, crude oil has financial property, whose price has prodigious impacts on other financial markets. For example, Narayan and Sharma [7–9] argue that stock return is related to crude oil price. Therefore, high prediction accuracy is quite helpful in asset allocation, risk management for investors and financial policy adjustment for policymakers. To this end, an accurate prediction for crude oil

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price benefits for stabilizing financial markets.

How to optimize prediction accuracy of crude oil price has been a hot issue in recent years. At present, the optimization is mainly carried out from two aspects. One is data processing, and the other is the selection of appropriate model. The former can effectively extract data features, and the latter can infer the tendency of target variable when selecting the suitable model.

1.1. Data processing method

The time series of crude oil price has many characteristics, such as linearity, nonlinearity, hysteresis, structural breaks and volatility, and it can be decomposed to generate sub-series or components with linearity, nonlinearity and fluctuation characteristics by decomposing algorithm [10]. Common data decomposing method include empirical mode decomposition (EMD) [11], ensemble empirical mode decomposition (EEMD) [2], complementary ensemble empirical mode decomposition (CEEMD) [10], complementary ensemble empirical mode decomposition with adaptive noise (CEEMDAN) [12], wavelet analysis [13], variational mode decomposition (VMD) [14] and so on. Decomposition techniques, like VMD, have significant advantages in mining the nonlinear and non-stationary properties hidden in time series, and the sequence can be decomposed into smoother and simpler components to make it more predictable [15]. For instance, Yu et al. [2] and Jiang et al. [12] adopt EEMD and CEEMDAN to decompose the original sequence into the several regular components and establish models respectively, whose results demonstrate that the model based on decomposed sequence has greater precision than other competing models. Abdollahi [10] testifies that CEEMD can improve the accuracy of crude oil price forecasting. Besides EMD and its advanced extensions, some scholars also attempt to conduct predictive analysis by wavelet analysis and VMD. Fortunately, Jammazi and Aloui [16] and Huang et al. [17] prove that these methods are proved to be useful and meaningful. In summary, these studies indicate that data processing plays a positive and remarkable role in optimizing prediction accuracy.

While data processing can help to improve the accuracy of predictions, there are some drawbacks in the existing researches, which is summarized as follows:

(1) Some algorithms are inherently flawed. For EMD, without an accurate mathematical model, its sensitivity to noise could affect the interpolation results and cause end effect and aliasing mode [18]. EEMD, CEEMD, and CEEMDAN perfect EMD by introducing noise in data processing, but they destroy the structural characteristics of original sequence. For wavelet analysis, wavelet basis selection and decomposition scale are currently facing challenges [19]. With respect to EMD, the perfected expansion of EMD and others, VMD without aliasing mode problem has distinct superiority in sequence denoising [20]. Importantly, this method can capture the trend and variability information of time series, so as to make the great improvement on the prediction accuracy [21]. Therefore, VMD is selected as the tool of data processing in the paper. Nevertheless, the decomposing number and penalty parameters of VMD are usually given through the previous experience [22] or the parameter of EMD algorithm [23].

(2) The above researches only use the regular data after denoising and ignore noise information identified by the decomposition method. In fact, the noise obtained by data processing method, as the part of sequence, could contain a lot of uncertainties but useful information, since it has impacts on the fluctuation features and range of sequence. To this end, it's necessary to quantify the noise information instead of discarding it in the prediction process.

1.2. The model for crude oil forecasting

Nowadays, scholars have done a lot of researches on crude oil price forecasting. To the best of our knowledge, relevant forecasting models can be classified into three categories, including econometric model,

artificial model, and hybrid model. It is common to see in the literature that a great many of econometric models are used to predict the price or volatility, including heterogeneous autoregressive-type (HAR-type) model [24], ARIMA model [25], vector autoregression (VAR) model [26], generalized autoregressive conditional heteroskedasticity type (GARCH-type) model [27], etc. Econometric models with explanatory parameters are simple and intuitive, and have a preponderance in describing the linearity of time series, but it is difficult to show nonlinear and non-stationary characteristics [10]. In particular, ARMA model is a vital method in the short-term prediction of stationary time series. In the past decade, benefiting from big data and development in computing power, artificial intelligence models have become more and more popular. These techniques are represented by long short-term memory (LSTM) model [28], convolutional neural network (CNN) model [29], back propagation (BP) model [30], ELM model [31], SVM model [32,33], and so on. Neural network models have evident advantages in fitting nonlinearity, but it also has some limitations such as overfitting and weak generalization ability. Especially, SVM model based on the statistical learning theory, has great superiority in small samples and nonlinear problems [34]. For the last type of forecasting model, hybrid models can be summarized into the direct hybrid model [35,36] and indirect hybrid model [2,37,38].

Numerous empirical evidence displays that the hybrid models have better predictive performance than the single models. For example, Safari and Davallou [35] establish a hybrid model based on the exponential smoothing model (ESM), autoregressive integrated moving average model (ARIMA), and nonlinear autoregressive neural network to make point prediction of oil price. Abdollahi and Ebrahimi [36] construct a hybrid model by the adaptive network-based fuzzy inference system (ANFIS) model, autoregressive fractionally integrated moving average (ARFIMA), and Markov-switching models. Safari and Davallou [35], Abdollahi and Ebrahimi [36] testify that the weighted hybrid model can improve the point prediction precision. Yu et al. [2] propose the hybrid models based on EEMD to predict crude oil price, E et al. [37] combine SVM, gated recurrent unit neural network, VMD and ICA to make prediction on energy price, and these researches find that the point prediction accuracy of hybrid models is more reliable and effective than other benchmark models. Besides, hybrid models such as [2,35,36], have attributes of econometric model and nonlinear model, which demonstrates that hybrid model has advantage in describe the data characteristics such as nonlinearity and linearity. These sufficient evidences motivate us to consider a hybrid model to predict crude oil price time series since it might not only have better prediction accuracy than the single model but depict the data features more precisely.

Hybrid model has gained widespread use. The direct hybrid model focuses on the original sequence, and it depends on the weight to adjust the prediction results. However, there is no unified standard for determining the weight value that affects the prediction accuracy directly. The indirect hybrid model concentrates on the decomposed subsequences, and different sub-models are employed to express data characteristics [2]. Comparing with the direct hybrid model, the indirect hybrid model based on data processing can not only describe the data characteristics well, but also don't need to determine the weight. Therefore, the indirect model is one of the focus in the study.

Many hybrid models have been developed on crude oil price forecasting, but there are still two weaknesses which are concluded below.

(1) It is worth to point out that the extant researches focus on the point prediction. However, comparing with point forecasting, interval forecasting provides an additional uncertainty assessment [39]. This is very important for economists and practitioners who pay attention to the crude oil market since a reliable interval estimation could guide them to make decisions with risk.

(2) It is lack of studies on online real-time prediction. Devpura et al. [40] verify that stock return predictability is time-varying, and crude oil price also has the same feature [41]. However, most of crude oil price forecasting methods pay attention to the offline prediction instead of

online real-time prediction, which can't embody the time-varying.

1.3. The research gap and current work

Going back to previous studies, there are some research gaps in crude oil price forecasting. (1) More and more researches prove the crucial role of VMD algorithm in modeling. However, with data updating, it is not objective to rely on empirical judgment or the number of EMD. So how to select the decomposing number and penalty parameters of VMD objectively is still worth to cause attention. (2) Thanks to the uncertainty of noise, it's difficult to quantify these information, which causes that few studies pay attention to the noise information after decomposing. (3) Existing hybrid models are rarely used for interval prediction and online real-time prediction. Nevertheless, interval prediction can help assess the fluctuation range of crude oil price in the future, and online real-time prediction is time-varying, which can provide a more accurate basis for the relevant crude oil practitioners than offline prediction.

For the sake of addressing the above shortcomings, corresponding solution methods are come up with. For solving VMD parameters, the intelligent algorithm can realize the determination of VMD parameters, but its fitness function needs to be carefully considered. Quoting maximum entropy [42] and envelope spectrum [43,44], average maximum envelope entropy (AMEE) is presented as the fitness function of intelligent algorithm. In view of noise, the original sequence is decomposed into several systematic sequences and noise sequence. Noise sequence is used to gain noise information matrix by Bootstrap sampling, and then the interval prediction results can be gained by combining this matrix and the point forecasting results. In this way, the problems of quantifying the noise uncertainty and interval prediction can both be addressed. If the model parameters can be automatically updated with data changing, online real-time forecasting will be achieved.

In summary, a novel method for crude oil price time series forecasting is put forward. Firstly, PSO algorithm with AMEE fitness function is adopted to get the decomposition number and punish parameters of VMD automatically. And we employ PSO-VMD to decompose the logarithm return (log-return) sequence of crude oil price into several sequences with regular information and noise. Secondly, given the merit of SVM model and ARMA model, these two models are selected as sub-models of regular sequences by Augmented Dikey-Fuller (ADF) test intelligently. The parameters of SVM model are determined by grid search [45], and the lag order of ARMA model is selected by minimum Akaike information criterion (AIC) principle [46]. In this way, a hybrid model (PVMD-SVM-ARMA model) is automatically established to make point prediction. Finally, PVMD-SVM-ARMA-B model added noise matrix is applied to make interval prediction. Beyond that, due to the automatic determination of model parameters, the presented models can be used for online real-time prediction in the empirical analysis.

There are three main contributions of this paper in theory, which is summarized as follows.

(1) A new method for crude oil price prediction is proposed. The proposed method has the intelligent properties, which reduces human intervention, and the operating process is simple and objective.

(2) A novel VMD parameter optimization method is presented. AMEE, as the fitness function of PSO algorithm, is proposed to obtain the parameters of VMD. The experiment results show that the parameters of VMD are changing with data information updating. Contrasting with the decomposition number of VMD determined by the experience or EMD, the presented method is more objective and scientific.

(3) A new method for quantifying data uncertainty is proposed. Noise information after decomposing the original sequence is quantified by Bootstrap resampling and interval prediction, which addresses the uncertainty of data sets. Meanwhile, the interval prediction results based on PVMD-SVM-ARMA-B model are proved to be reasonable and effective.

In terms of application, the contributions are concluded as follows.

(1) The proposed method has good prediction accuracy. According to this method, we can know not only the fluctuation trend of crude oil price but also its fluctuation range. These provide the reference for crude oil investment and oil trade. For example, when crude oil price is predicted to rise, it offers an investment opportunity for investors, and an early inflation warning for oil-importing countries.

(2) The proposed method has high flexibility, which can be used for both point prediction and interval prediction. What's more, their prediction results are both demonstrated to be reliable and valid.

(3) The proposed method can be used for online real-time prediction. PVMD-SVM-ARMA model and PVMD-SVM-ARMA-B model are used for online real-time point and interval prediction respectively. And the prediction results show that they are with great forecasting precision and reliability.

(4) PVMD-SVM-ARMA model has great prediction performance and excellent robustness. According to the loss functions, DM test and MCS test, it can be known that PVMD-SVM-ARMA model outperforms other models significantly in the different data sets. In addition, PVMD-SVM-ARMA model embodies the stable prediction performance in the data sets with different frequencies and different periods, which verifies that PVMD-SVM-ARMA model has relatively reliable robustness.

(5) PVMD-SVM-ARMA model is generalized in crude oil price prediction. This model performs well in expansion and recession periods. In addition, it does a good job in predicting positive and negative returns even if structural breaks occur, which means that it can describe the rise and fall of crude oil prices. In the meantime, it also proves that PVMD-SVM-ARMA model is also effective even though the global shock like COVID-19 occurs.

(6) PVMD-SVM-ARMA-B model can predict the fluctuation range and characteristics of crude oil log-return well. As can be seen from Sub-section 4.2.4, PVMD-SVM-ARMA-B model quantifies the uncertainty of noise information, and its results can provide a reference of crude oil price fluctuation range and tendency.

The rest structure of the paper is organized as follows. Section 2 introduces the used algorithms and models, including VMD, SVM model, ARMA model, and the proposed method. Section 3 states the evaluation indicators. Section 4 is the empirical application. Section 5 is the conclusion, and the last is the future work.

2. Method

This section introduces VMD algorithm, the basic model and presented method in detail. The basic model includes ARMA model and SVM model, which are adopted to fit subsequences and accomplish prediction. The presented methods include PSO-VMD algorithm and the proposed model, namely PVMD-SVM-ARMA model and PVMD-SVM-ARMA-B model.

2.1. Variational mode decomposition

VMD proposed by Konstantin Dragomiretskiy and Dominique Zosso, is an adaptive and completely non-recursive decomposition method, which has advantages in dealing with noise and nonlinear sequence [20]. In VMD algorithm, the quadratic penalty term and Lagrange multiplier are applied to transform the constrained problem into a non-constrained problem, and then the alternative direction multiplier method is used to resolve the nonlinear problem and calculate all the mode functions. The constrained problem is written as

$$\min \left\{ \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} s.t. \sum_{k=1}^K u_k = f, \quad (1)$$

where u_k is the k th intrinsic mode function, ω_k is corresponding center frequency, and f is input signal. The non-constrained problem is shown as

$$L(\{u_k\}, \{\omega_k\}, \lambda) = \alpha \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 + \left\| f(t) - \sum_{k=1}^K u_k(t) \right\|_2^2 + \lambda \left(f(t) - \sum_{k=1}^K u_k(t) \right) > 0. \quad (2)$$

where $\lambda(t)$ is Lagrange multiplier, and α is penalty factor. Parameters $\hat{u}_k$, $\hat{\omega}_k$ and $\hat{\lambda}$ are updated alternately, and the updating process can be specified by

$$\hat{u}_k^{n+1}(\omega) = \frac{\hat{f}(\omega) - \sum_{i < k} \hat{u}_i^{n+1}(\omega) - \sum_{i > k} \hat{u}_i^n(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k^n)^2}, \quad (3)$$

$$\omega_k^{n+1} = \frac{\int_0^\infty \omega |u_k^{n+1}(\omega)|^2 d\omega}{\int_0^\infty |u_k^{n+1}(\omega)|^2 d\omega}, \quad (4)$$

$$\hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \tau \left(\hat{f}(\omega) - \sum_k \hat{u}_k^{n+1}(\omega) \right). \quad (5)$$

To show the process of VMD more straightforwardly, we provide the pseudocode as displayed in Algorithm 1 of B, in which ζ is the tolerance of convergence criterion.

2.2. Basic model

2.2.1. Autoregressive moving average model

ARMA model proposed by American statistician Box and British statistician Jenkins is a time-series analysis model. Its expression is expressed by

$$x_t = \varphi_0 + \sum_{i=1}^p \varphi_i x_{t-i} + e_t - \sum_{j=1}^q \theta_j e_{t-j}, \quad (6)$$

where φ_i and p are the coefficient and order of autoregressive term respectively, θ_j and q are the coefficient and order of moving average term separately. When $p = 0$, model degenerates to be moving average model with q lags, namely MA(q) model, and when $q = 0$, model degenerates to be autoregressive model, namely AR(p) model. Because ARMA model might appear wrong statistical inference for non-stationary data, it is only effective for stationary time series modeling. Besides, ARMA model tends to be stable and lose prediction significance for long-term predictions.

2.2.2. Support vector machine

Taking the minimum structural risk as the principle, SVM model [47] with strong generalization ability, can effectively solve nonlinear problems and fulfill small sample prediction. Its basic idea is mapping the input data to the high-dimensional feature space by nonlinear transformation, so as to transform the nonlinear problem of low-dimensional space into the linear problem of high-dimensional space. The expression of support vector regression is written as

$$f(x) = v * \phi(x) + b, \quad (7)$$

in which, $f(x)$ is the forecasting function, v is the weight vector, $\phi(x)$ is mapping function, and b is the offset constant. v and b can be estimated according to minimizing the structural risk, and the optimizing problem can be written as

$$\begin{aligned} \min z = & \frac{1}{2} \|v\|^2 + c \sum_{i=1}^n (\xi_i + \xi_i^*) \\ \text{s.t.} \quad & \begin{cases} y_i - v^T \phi(x_i) - b \leq \varepsilon + \xi_i \\ -y_i + v^T \phi(x_i) + b \leq \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0, i = 1, 2, \dots, n \end{cases} \end{aligned} \quad (8)$$

where ξ_i and ξ_i^* is relaxation factor, c is penalty factor applied to control the punishment degree beyond the sample, and ε is the given parameter. The goal of $c \sum_{i=1}^n (\xi_i + \xi_i^*)$ is reducing error. In order to solve Eq. 8, Lagrange multiplier β_i, β_i^* , and kernel function $K(x_i, x_j)$ are introduced, and its dual form can be obtained as

$$\begin{aligned} \max w = & \sum_{i=1}^n y_i (\beta_i - \beta_i^*) - \varepsilon \sum_{i=1}^n (\beta_i + \beta_i^*) - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (\beta_i - \beta_i^*) (\beta_j - \beta_j^*) K(x_i, x_j) \\ \text{s.t.} \quad & \begin{cases} \sum_{i=1}^n (\beta_i - \beta_i^*) = 0 \\ 0 \leq \beta_i, \beta_i^* \leq c, i = 1, 2, \dots, n \end{cases} \end{aligned} \quad (9)$$

According to Eq. 9, we can get

$$v = \sum_{i=1}^n (\beta_i - \beta_i^*) \phi(x_i). \quad (10)$$

When the kernel function is given, the regression function is

$$f(x) = \sum_{i=1}^n (\beta_i - \beta_i^*) K(x, x_i) + b. \quad (11)$$

2.3. The proposed method

2.3.1. PSO-VMD algorithm

In VMD algorithm, the decomposition number and the penalty parameters are of great concern. Information entropy [48] is a measuring indicator of the system regularity. Therefore, the maximum information entropy (also known as maximum entropy) is introduced to excavate the potential and ordered information from the sequence. Based on maximum entropy and envelope spectrum, AMEE is introduced as the objective function to excavate regular feature information of time series. That is, AMEE is used as the fitness function of PSO algorithm to optimize the parameters of VMD algorithm, which is called PSO-VMD algorithm. The fitness function is calculated as follows.

Step 1: Sequence namely signal is made Hilbert transformation. That is $U = \text{hilbert}(u)$, where u is the input sequence, and U is the output sequence obtained by Hilbert transformation.

Step 2: Envelope signal is extracted, that is $s_i = \sqrt{U_i^2 + u_i^2}$, where U_i and u_i are the i th output and input sequence in Hilbert transformation, and s_i is the corresponding envelope signal.

Step 3: Envelope spectrum is obtained by the absolute of Fourier transformation, namely $F_i = \text{abs}(FFT(s_i))$, in which F_i is the envelope spectrum.

Step 4: Envelope spectrum is normalized by $Se_i = F_i / \sum_{i=1}^K F_i$, where K is the number of sequence, and Se_i is the standardized envelope spectrum.

Step 5: Envelope entropy is calculated by $E_i = -\sum_{j=1}^l Se_{ij} * \log_2 Se_{ij}$, in which, l is the length of Se_i .

Step 6: The fitness function is obtained by AMEE, which is

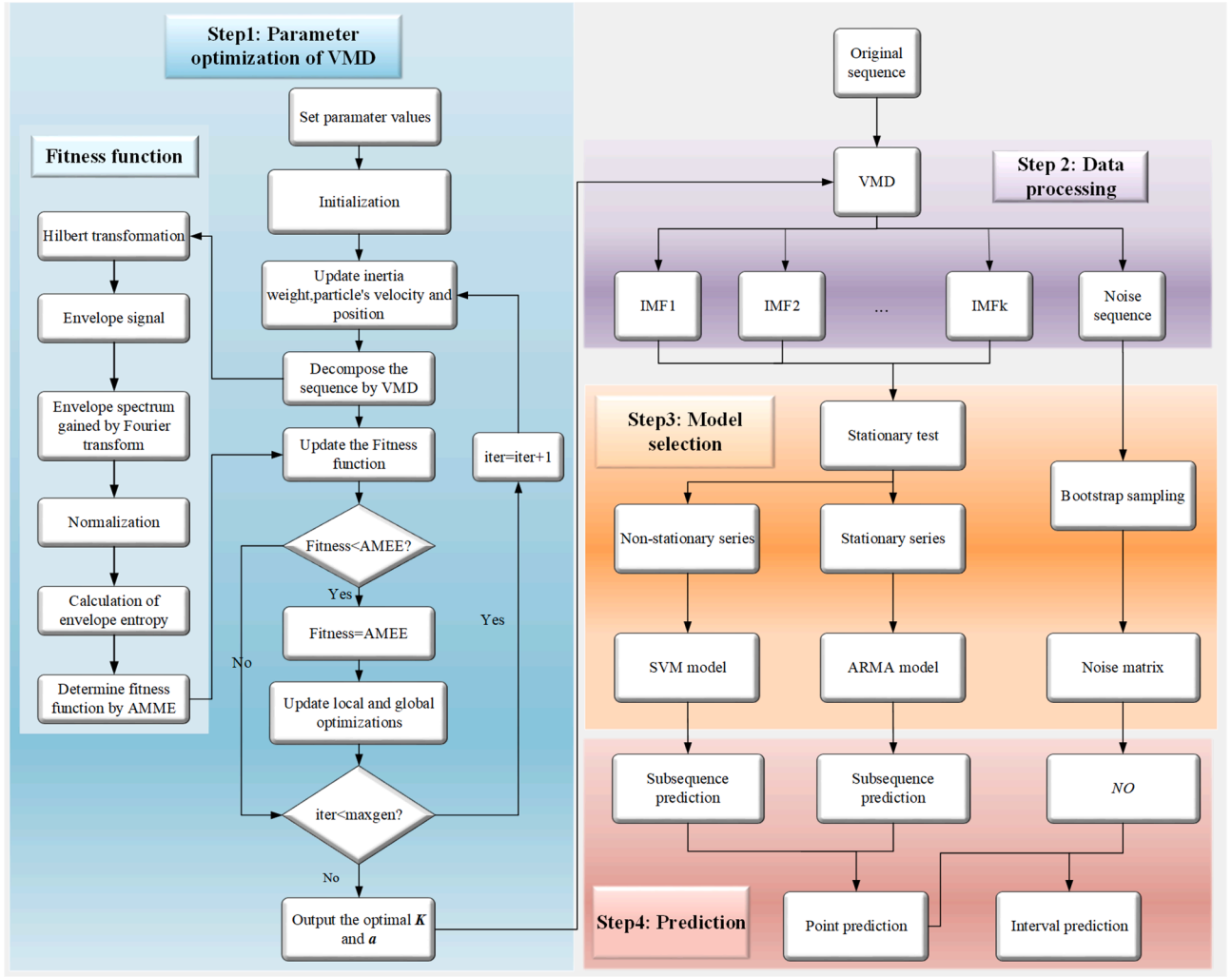


Fig. 1. The flow chart of proposed model.

$$AMEE = \arg \max_{(K, \alpha)} \left(\frac{1}{K} \sum_{i=1}^K E_i \right)_{size}, \quad (12)$$

in which, *size* is the population size.

Particle swarm optimization (PSO) algorithm is an intelligent optimization algorithm proposed by J. Kennedy and R. C. Eberhart, whose basic idea is to seek the optimal solution according to the individual cooperation and group information sharing. In PSO algorithm, each solution is a particle with two properties, namely position and velocity. Depending on the fitness function, position and velocity can be updated. With genetic algorithm compared, PSO algorithm with the simple rule is easier to implement, and has superiority of high precision and fast convergence.

Presetting AMEE as the fitness function, parameters *K* and *α* can be got by PSO-VMD algorithm. Set $P = [P_1, P_2]$, $V = [V_1, V_2]$, $P_{\max} = [P_{\max1}, P_{\max2}]$, $P_{\min} = [P_{\min1}, P_{\min2}]$, $V_{\max} = [V_{\max1}, V_{\max2}]$ and $V_{\min} = [V_{\min1}, V_{\min2}]$, hence the steps of VMD-PSO algorithm are as follows.

Step 1: Set the initial parameter values. The initial values are given, including inertia weight range $[\omega_{\min}, \omega_{\max}]$, acceleration factor c_1, c_2 , population size *size*, maximum iteration number *maxgen*, maximum and minimum of particles $P_{\max,i}, P_{\min,i}$, maximum and minimum velocity $V_{\max,i}, V_{\min,i}$ in which $i = 1, 2$.

Step 2: Initialization. Position P_i and velocity V_i are initialized randomly. According to AMEE, the fitness function *Fitness*, the individual optimal position $pbest_i$, the individual optimal fitness $fbest_i$, the

global optimal position $Pbest_i$ and global optimal fitness $Fbest_i$ can be calculated initially.

Step 3: Update inertia weight by

$$\omega = \omega_{\max} - \frac{iter * (\omega_{\max} - \omega_{\min})}{maxgen}, \quad (13)$$

where *iter* is the current iteration number, and *maxgen* is the maximum iteration number.

Step 4: Update the particle position and velocity by

$$\begin{aligned} V_i &= \omega^* V_i + c_1 * rand() * (pbest_i - P_i) + c_2 * rand() * (Pbest - P_i) \\ P_i &= P_i + V_i \end{aligned} \quad (14)$$

Step 5: Update the fitness function *Fitness*, individual and global optimal position $pbest_i, Pbest_i$ by AMEE.

Step 6: Judge whether the iteration number is more than *maxgen*. If so, the algorithm ends, if not, then returns **Step 3**.

According to the above steps, the pseudocode of PSO-VMD algorithm is written as Algorithm 2 of B. For convenience, PSO-VMD algorithm in the proposed models is simplified as PVMD.

2.3.2. Proposed model

The existing models neither quantify the noise information nor pay attention to interval prediction. Point prediction can provide direct and effective reference information for decision makers, but interval prediction can estimate uncertainty information that cannot be quantified

by point prediction [49]. Therefore, both interval prediction and point prediction are of great significance in practical application. Based on the above analysis, this paper proposes a new method for crude oil price forecasting. This method can not only achieve point prediction but also combine Bootstrap and interval prediction to quantify the uncertainties of noise. The proposed method is mainly divided into four steps, namely VMD parameter optimization, data processing, model selection, and prediction. The prediction flow chart is shown in Fig. 1.

Step 1: Parameter optimization of VMD.

PSO is used to optimize the parameters of VMD. Firstly, the envelope entropy of subsequence is calculated by Hilbert transformation, Fourier transform, and information entropy. Then the fitness function is determined according to the AMEE. Finally, with the fitness function updating, the optimal K and α can be determined. According to the setting of K in the existing literature [50,51], the range of K is set as [5, 20]. Referring to [42], the range of α is set as [1000, 10000]. Specific implementation steps of parameter optimization are given in Algorithm 2.

Step 2: Data processing.

On ground of the optimal parameters K and α , original sequence is decomposed by Algorithm 1 to obtain several intrinsic mode functions (IMFs) with the regular information and noise sequence, which can weaken the random volatility of original sequence and improve the sequence predictability.

Step 3: Model selection.

Different processing measures are taken for IMFs and noise sequences. Given the stationary features of IMFs, the sequence is identified as stationary sequence and non-stationary sequence by ADF test. ARMA model can be built for stationary sequences. Taking small sample and nonlinearity characteristic into consideration, SVM model can be built for non-stationary sequences. Then the computer automatically selects SVM model or ARMA model by ADF test result. In the meantime, the p and q of ARMA(p, q) model is automatically chosen according to minimum AIC principle, and grid search method and linear kernel function are applied to determine the parameters of SVM model intelligently. In view of noise sequence, Bootstrap resampling technique is used to ensure the integrity of sequence information.

Step 4: Prediction.

Step 4-1: Point prediction. The hybrid model for point prediction is built on ground of IMFs, which is called PVMD-SVM-ARMA model.

On basis of the model established in Step 3, IMFs are forecasted by rolling prediction of one step ahead, and the prediction scheme is described in the C.2 specifically. The point prediction results $\hat{x}$ can be got by combining the predicted value $\hat{I}_i$ of IMFs. The rolling prediction is performed m times to obtain the forecasting set.

Step 4-2: Interval prediction. Adding the noise matrix into the point prediction results, the interval prediction results are obtained. Hereinafter, the hybrid model with noise information is called PVMD-SVM-ARMA-B model.

The detailed process of PVMD-SVM-ARMA, PVMD-SVM-ARMA-B and PVMD-SVM-ARMA-Bmax model is shown in C.1, and the flow chart of proposed models are shown in Fig. 1. According to the above modeling steps, a new predictor with online real-time prediction is constructed, which is shown in Algorithm 3 of B. That is, the model will update with the data changing in predictor.

3. Evaluation principle

This section is classified into two subsections, namely the evaluation of point prediction and the evaluation of interval prediction.

3.1. The evaluation of point prediction

There are three commonly used indicators of point prediction in the machine learning model and econometric model, including mean square

error (MSE), mean absolute error (MAE), and Theil inequality coefficient (TIC) [52]. MSE is defined as

$$MSE = \frac{1}{N} \sum_{t=1}^N (\hat{x}_t - x_t)^2. \quad (15)$$

MAE is specified by

$$MAE = \frac{1}{N} \sum_{t=1}^N |\hat{x}_t - x_t|. \quad (16)$$

TIC is calculated by

$$TIC = \frac{\sqrt{\frac{1}{N} \sum_{t=1}^N (\hat{x}_t - x_t)^2}}{\sqrt{\frac{1}{N} \sum_{t=1}^N x_t^2} + \sqrt{\frac{1}{N} \sum_{t=1}^N \hat{x}_t^2}}, \quad (17)$$

where x_t is the actual value, $\hat{x}_t$ is the corresponding prediction value, and N is the out-of-sample size. The less MSE, MAE and TIC are, the higher the prediction accuracy is.

For investors, they will be more interested in whether the return is positive or negative since its sign can help investors identify investment points. That is also the reason why we will pay more attention to the plus-minus sign of return rather than its rise and fall. Therefore, in the evaluation of point prediction, the common consistency indicator of rise and fall is replaced by the sign consistency (SC). SC is given by

$$SC = \frac{\sum_{t=1}^N \lambda_t}{N}. \quad (18)$$

In Eq. 18, $\lambda_t = \begin{cases} 1, & \text{if the signs of } \hat{x}_t \text{ and } x_t \text{ are the same} \\ 0, & \text{if the signs of } \hat{x}_t \text{ and } x_t \text{ are not the same} \end{cases}$, the range of SC is [0, 1]. SC is greater, and the change of predicted value is more consistent with that of original sequence.

In order to prove the superiority of proposed model, the predicted results are tested by Diebold Mariano (DM) test [53] and model confidence set (MCS) test [54]. Based on loss function, DM test is used to compare whether there is significant difference between two predicted sequences from the view of statistics. The test statistics are expressed by

$$dm = \frac{\bar{d}}{\sqrt{\frac{2\pi f_d(0)}{N}}} \sim N\left(0, 1\right), \quad (19)$$

in which, $\bar{d} = \frac{1}{T} \sum_{t=1}^T d_{AB,t}$, $d_{AB,t} = g(e_{A,t}) - g(e_{B,t})$, $\gamma_d(\tau) = E[(d_t - \bar{d})(d_{t-\tau} - \bar{d})] = E(d_t)$, $f_d(0) = \frac{1}{2\pi} \sum_{\tau=-\infty}^{\infty} \gamma_d(\tau)$. $g()$ is the loss function corresponding to the error, $e_{A,t}$ and $e_{B,t}$ are the error sequences of model A and model B respectively.

MCS test does not need to set the benchmark model, and it only needs to compare multiple models through equivalence test and elimination rule. Then, the optimal model set can be obtained in a given confidence level. Taking loss function (MAE and MSE) as the evaluation basis, MCS test combined with Bootstrap method to obtain the two statistics, namely TR and TMAX. The calculating formula of TR is

$$TR = \max_{i,j \in \mathcal{M}} \frac{|\bar{d}_{ij}|}{\sqrt{\hat{v}(\bar{d}_{ij})}}. \quad (20)$$

TMAX is calculated by

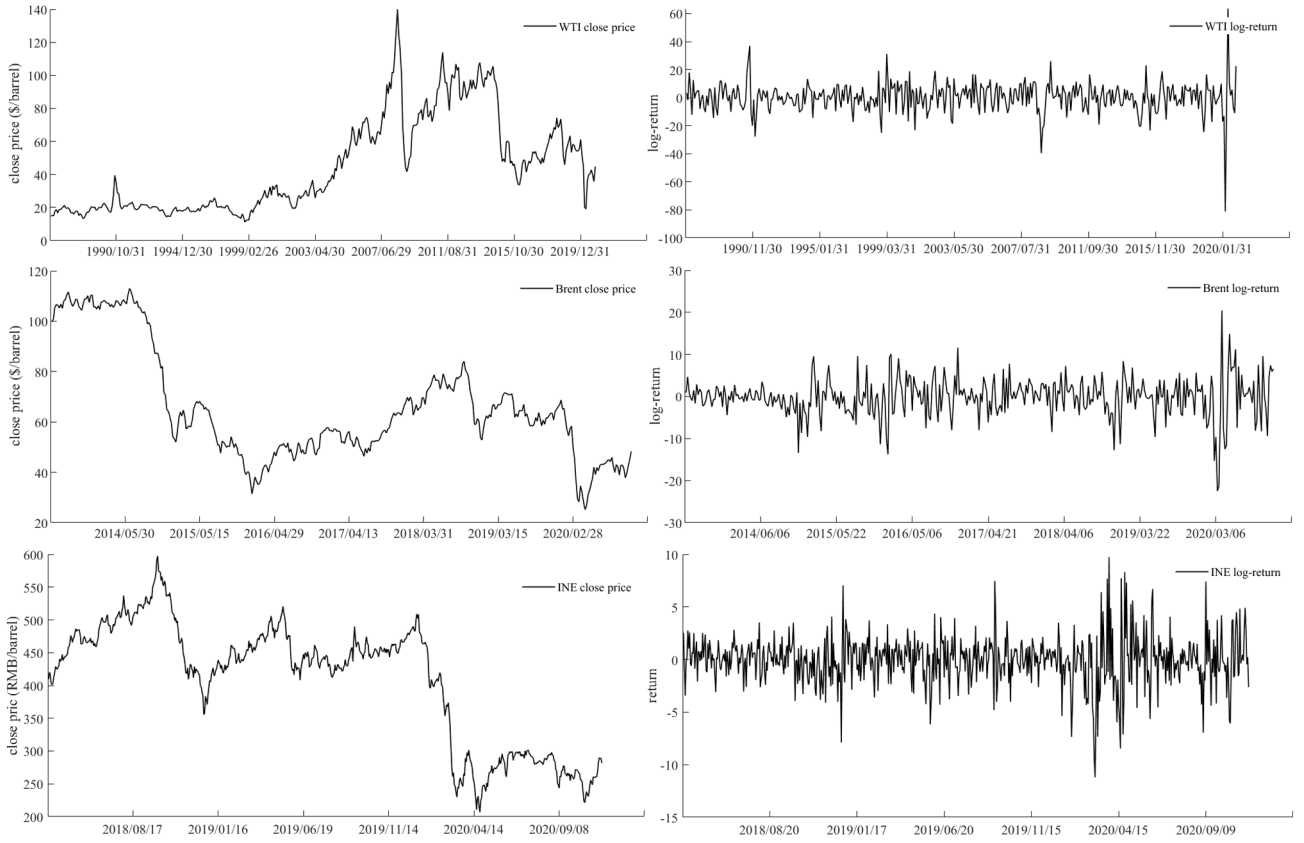


Fig. 2. Crude oil price and the corresponding log-return in WTI, Brent and INE.

$$TMAX = \max_{i,j \in \mathcal{H}} \frac{\bar{d}_{i,j}}{\sqrt{\hat{v}(\bar{d}_{i,j})}} \quad (21)$$

where $\bar{d}_{ij} = \bar{d}, \bar{d}_{i,j} = \frac{1}{M} \sum_{j \in \mathcal{H}} \bar{d}_{ij}$, $\hat{v}(\bar{d}_{ij})$ and $\hat{v}(\bar{d}_{i,j})$ are the estimators of $\text{var}(\bar{d}_{ij})$ and $\text{var}(\bar{d}_{i,j})$ respectively.

3.2. The evaluation of interval prediction

General evaluation indexes of interval prediction include the coverage rate r_{cov} of true value and the interval diameter D , where r_{cov} and D represent the reliability and sensibility of interval separately. r_{cov} is expressed by

$$r_{cov} = \frac{1}{N} \sum_{t=1}^N \gamma_t, \quad (22)$$

in which, $\gamma_t = \begin{cases} 0, & x_t \notin \tilde{x}_t \\ 1, & x_t \in \tilde{x}_t \end{cases}$, $\tilde{x}_{t,u}$ and $\tilde{x}_{t,l}$ are the upper and lower bound of interval respectively, and $\tilde{x}_t$ is the predictive interval of actual value x_t . The higher r_{cov} is, the more reliable prediction results are. Interval diameter D is calculated by

$$D = \frac{1}{N} \sum_{t=1}^N (\tilde{x}_{t,u} - \tilde{x}_{t,l}). \quad (23)$$

The smaller interval diameter is, the stronger sensitivity of corresponding prediction is, and the greater reference value for future information is.

r_{cov} and D can't evaluate the fluctuation characteristics of sequence, hence the fluctuation consistency C is introduced to describe the accuracy of fluctuation features, which is estimated by

$$C_l = \frac{\sum_{t=1}^{N-1} S_{t,l}}{N-1}, \quad (24)$$

$$C_u = \frac{\sum_{t=1}^{N-1} S_{t,u}}{N-1}$$

in which, $S_{t,l(u)} = \begin{cases} 1, & (\tilde{x}_{t+1,l(u)} - \tilde{x}_{t,l(u)}) * (x_{t+1} - x_t) > 0 \\ 0, & \text{otherwise} \end{cases}$, C fluctuates in $[0, 1]$. The larger C is, the higher effectiveness of interval prediction is.

4. Empirical application

Empirical research for crude oil price log-return forecasting is conducted to verify the feasibility and validity of proposed method. To do this, we focus on three types of crude oil futures products, including WTI traded in New York Mercantile Exchange, Brent traded in London Intercontinental Exchange, and INE traded in Shanghai International Energy Exchange. The experiment in this section is accomplished by Matlab and R. Section 4.1 makes a description of WTI, Brent, and INE crude oil price. In Section 4.2, we discuss the online prediction results in four subsections, including parameter optimization of VMD, model comparison, the analysis of point prediction, and the analysis of interval prediction.

4.1. Data description

We collect closing prices of WTI, Brent, and INE crude oil futures from the Choice database of EastMoney. Since WTI, Brent, and INE started trading in 1986, 2013, and 2018 respectively, our sample periods span from September 30, 1986 to November 30, 2020, from June 21, 2013 to November 27, 2020, and from March 29, 2018 to November 30,

Table 1
Statistical description of crude oil price and log-return.

	mean	St.Dev	max	min	skewness	kurtosis	J-B (<i>p</i>)	ADF (<i>p</i>)
WTI close price	44.771	28.980	140.000	11.220	0.853	2.618	0.001	0.362
WTI log-return	0.270	10.529	63.491	-81.170	-0.709	14.605	0.001	0.001
Brent close price	65.686	21.953	113.050	25.180	0.787	2.656	0.001	0.127
Brent log-return	-0.187	4.676	20.420	-22.452	-0.470	6.348	0.001	0.001
INE close price	408.367	92.571	597.300	206.600	-0.581	2.076	0.001	0.377
INE log-return	-0.058	2.342	9.732	-11.203	-0.165	5.813	0.001	0.001

Note: This table reports the descriptive statistics of crude oil price and its log-return. St.D, J-B (*p*) and ADF (*p*) represent the standard deviation, the *p* value of Jarque–Bera test and Augmented Dikey–Fuller unit root test. The null hypothesis of J-B test is that the variables obey the standard normal. The null hypothesis of ADF test is that the sequence does not have unit roots.

Table 2
VMD parameters optimized by PSO algorithm with AMEE fitness fuction.

	WTI			-	Brent			-	INE		
	Stage 1	Stage 2	Stage3		Stage 1	Stage 2	Stage3		Stage 1	Stage 2	Stage3
K	16	15	17		19	15	19		5	20	19
alpha	9420	9775	2633		9598	9034	8406		8593	9136	8332

Note: This table reports the VMD parameters optimized by improved PSO algorithm. *K* is the decomposed number, and α is the penalty parameters.

2020 respectively. In addition to the different categories of crude oil futures products, we further consider data at different frequencies. More specifically, monthly WTI, weekly Brent, and daily INE are used to analyze the effectiveness of our proposed model.¹ For the convenience of display and calculation, we calculate log-return of crude oil by expanding it a hundredfold, which is specified by

$$r_t = 100^* (\log p_t - \log p_{t-1}), \quad (25)$$

in which, p_t is closing price of WTI, Brent or INE crude oil at *t* moment.

Fig. 2 shows crude oil price series and its log-return series, and Table 1 shows its statistical characteristics of crude oil price and its log-return. We find that the mean values of WTI log-return is positive while that of Brent and INE log-return are negative. Besides, kurtosis, skewness, and J-B test results expound that prices and log-return series of all crude oil products deviate from the normal distribution. The last column shows the *p* values of ADF test, which suggests that the WTI, Brent, and INE crude oil price series are non-stationary, but their log-return series are stationary. Furthermore, Fig. 2 shows the dynamic fluctuation of crude oil price series and its log-return series. We observe that both price series and log-returns series have evident nonlinear characteristics. It is well-known that econometric models have advantages in describing stationary features, but they cannot accurately fit nonlinear characteristics. Given these two features, log-return is decomposed into several smoother sub-series by PVMD to build the hybrid model based on ARMA and SVM model. Hence the proposed method is rational for log-return prediction and helpful to reflect the hidden fluctuation characteristics.

4.2. Online real-time prediction

Time selection and sample data frequency will affect the original prediction accuracy [55]. Therefore, we use online updating of data and models to realize the prediction at different frequency data. When ARMA model is used for prediction, the test set of 3% to 8% is reasonable [56,57]. Therefore, based on the Algorithm 3, online real-time predictions of three phases can be accomplished, and the updating window of WTI, Brent and INE are two years, nearly four months and six weeks respectively. For WTI, test set of Stage 1 is from December 31, 2014 to November 30, 2016, these of Stage 2 and Stage 3 are from December 30,

2016 to November 30, 2018, and from December 31, 2018 to November 30, 2020 respectively. For Brent, test sets of three stages are from November 22, 2019 to March 20, 2020, from March 27, 2020 to July 24, 2020, and from July 31, 2020 to November 27, 2020 severally. For INE, test sets of three stages are from July 20, 2020 to August 28, 2020, from August 31, 2020 to October 19, 2020, and from October 20, 2020 to November 30, 2020 severally.

4.2.1. The optimized parameters of VMD

According to the online prediction scheme of Fig. C.5, the training set and model are updated with historical information changing. Table 2 displays the parameter optimizing results of VMD based on Algorithm 2. As shown in Table 2, VMD parameters will be updated as the data changes. Updating data will change the internal information characteristic, which makes information entropy changed, since the parameters of VMD will change accordingly. In conclusion, PSO-VMD algorithm is effective.

4.2.2. Model comparison

In order to show whether the proposed model is dominant, we consider a series of single models and hybrid models as competing models. More specifically, based on ARMA model, SVM model and ELM model, several hybrid models are established by combining with VMD, EMD, EEMD, CEEMD, and CEEMDAN separately. For the sake of description hereinafter, Models 1–12 represent PVMD-SVM-ARMA model, PVMD-SVM-ARMA-Bmax² model, ARMA model, SVM model, EMD-SVM-ARMA model, EEMD-SVM-ARMA model, CEEMD-SVM-ARMA model, CEEMDAN-SVM-ARMA model, EMD-ELM-ARMA model, EEMD-ELM-ARMA model, CEEMD-ELM-ARMA model, CEEMDAN-ELM-ARMA model respectively.³

As can be seen from Table 3, point prediction evaluation results during Stages 1–3 of Models 1–12 are reported. Firstly, we find that MSEs, MAEs, and TICs of the PVMD-SVM-ARMA model are all less than these of other models from Stage 1 to Stage 3 in three different data sets. Meanwhile, comparing with other models, MAEs and MSEs of PVMD-

² Before Bootstrap sampling, noise sequence has been carried out autocorrelation analysis, which shows that it has random characteristics.

³ Gaussian white noise will be added in the decomposition process of EEMD, CEEMD and CEEMDAN, which might lead to the inconsistency of decomposition results, and thus contribute to the instability of prediction results. We carry out five experiments to get the average prediction values of hybrid models based on EEMD, CEEMD and CEEMDAN, so as to keep their stability.

¹ This frequency setting is based on the periods of crude oil futures. For example, WTI has the longest trading period and thus monthly frequency is selected.

Table 3
The point prediction evaluation of three stages.

		WTI					Brent					INE				
		MSE	MAE	SC	TIC	TIME(s)	MSE	MAE	SC	TIC	TIME(s)	MSE	MAE	SC	TIC	TIME(s)
Stage 1	Model 1	29.916	4.331	83.3%	0.290	2002.422	24.972	3.190	88.9%	0.347	1437.422	1.146	0.848	66.7%	0.600	983.922
	Model 2	71.906	6.435	75.0%	0.433	2006.531	38.701	4.101	83.3%	0.453	1443.141	12.791	2.477	33.3%	0.807	990.141
	Model 3	123.768	8.422	54.2%	0.810	38.109	84.005	6.436	66.7%	0.973	48.156	1.277	0.877	40.0%	0.956	37.859
	Model 4	131.321	8.856	58.3%	0.929	10.734	85.281	6.418	55.6%	0.974	10.078	1.119	0.823	63.3%	0.761	33.047
	Model 5	115.199	8.476	50.0%	0.769	203.375	54.360	5.570	72.2%	0.606	157.313	1.277	0.881	50.0%	0.767	261.750
	Model 6	104.238	7.924	50.0%	0.672	997.844	55.067	5.340	72.2%	0.556	840.906	1.396	0.958	46.7%	0.729	1441.516
	Model 7	101.122	7.731	54.2%	0.571	1845.063	47.711	5.081	72.2%	0.527	1063.859	1.168	0.870	63.3%	0.675	1888.188
	Model 8	105.548	7.817	50.0%	0.686	967.719	49.671	5.205	72.2%	0.540	816.234	1.458	0.979	40.0%	0.742	1355.500
	Model 9	126.964	8.779	58.3%	0.724	179.359	116.737	8.179	61.1%	0.634	132.563	4.644	1.713	56.7%	0.680	137.344
	Model 10	122.813	8.898	54.2%	0.748	755.578	68.711	6.126	66.7%	0.739	684.234	6.098	2.057	50.0%	0.775	670.000
	Model 11	150.954	10.400	66.7%	0.582	1730.719	221.973	10.326	50.0%	0.722	727.438	1.255	0.899	60.0%	0.770	1190.156
	Model 12	119.485	8.840	54.2%	0.738	778.375	60.126	5.801	72.2%	0.652	681.297	1.440	0.941	56.7%	0.708	655.500
Stage 2	Model 1	15.364	3.017	87.5%	0.299	1330.703	12.801	2.988	88.9%	0.223	1000.828	1.187	0.823	90.0%	0.221	1986.109
	Model 2	61.687	6.367	62.5%	0.596	1334.750	27.518	4.098	83.3%	0.339	1006.219	3.012	1.334	73.3%	0.363	1992.156
	Model 3	62.739	6.145	54.2%	0.788	39.703	91.200	7.641	44.4%	0.788	45.359	8.136	2.152	46.7%	0.927	45.891
	Model 4	64.852	6.452	54.2%	0.897	10.625	80.087	7.359	38.9%	0.993	10.234	8.212	2.218	43.3%	0.882	32.953
	Model 5	54.420	5.444	62.5%	0.622	193.984	78.895	6.652	61.1%	0.682	154.641	6.996	1.965	63.3%	0.633	267.000
	Model 6	64.087	6.253	45.8%	0.706	980.734	78.395	7.312	55.6%	0.759	976.203	7.029	1.962	60.0%	0.647	1434.313
	Model 7	75.846	6.828	58.3%	0.668	1733.328	67.300	6.655	61.1%	0.672	1455.297	7.596	1.987	63.3%	0.698	1752.656
	Model 8	61.641	6.340	45.8%	0.685	945.742	80.311	7.521	50.0%	0.765	914.195	7.047	1.952	56.7%	0.661	1329.391
	Model 9	62.111	6.477	54.2%	0.619	182.125	83.571	7.578	44.4%	0.799	111.188	7.907	2.082	73.3%	0.618	141.250
	Model 10	74.823	6.780	50.0%	0.711	746.203	75.798	7.315	44.4%	0.810	750.516	6.833	1.969	50.0%	0.669	684.563
	Model 11	452.517	16.807	58.3%	0.812	1604.109	70.868	6.991	55.6%	0.714	1411.688	8.110	1.998	63.3%	0.696	886.406
	Model 12	212.081	9.057	50.0%	0.634	685.109	80.397	7.493	50.0%	0.790	806.016	11.981	2.662	53.3%	0.676	672.656
Stage 3	Model 1	37.143	3.797	87.5%	0.145	1205.672	3.257	1.351	94.4%	0.169	1255.203	1.028	0.832	83.3%	0.201	1856.250
	Model 2	53.901	4.903	91.7%	0.177	1209.750	10.634	2.685	77.8%	0.313	1260.578	2.595	1.223	80.0%	0.314	1862.281
	Model 3	535.226	13.997	62.5%	0.885	39.469	35.738	5.048	38.9%	0.932	35.906	8.821	2.345	30.0%	0.939	41.063
	Model 4	540.181	13.832	54.2%	0.947	10.906	32.818	4.797	61.1%	0.968	10.109	7.980	2.155	60.0%	0.973	32.641
	Model 5	503.206	13.477	62.5%	0.779	213.000	29.300	3.918	61.1%	0.627	233.703	6.949	2.186	56.7%	0.580	265.234
	Model 6	512.870	13.873	66.7%	0.775	1326.016	31.293	4.321	50.0%	0.657	1174.219	7.482	2.242	53.3%	0.605	1437.547
	Model 7	544.065	15.259	62.5%	0.755	1564.188	50.010	5.348	38.9%	0.710	1346.469	7.398	2.294	50.0%	0.691	1757.891
	Model 8	506.911	13.786	66.7%	0.791	1193.336	34.105	4.388	66.7%	0.665	978.391	8.331	2.295	56.7%	0.640	1309.594
	Model 9	513.399	13.576	66.7%	0.787	190.172	34.205	4.509	50.0%	0.598	229.109	12.101	2.872	53.3%	0.664	132.438
	Model 10	524.096	13.538	62.5%	0.799	1329.594	31.895	4.527	72.2%	0.751	1013.109	6.990	2.162	53.3%	0.558	668.609
	Model 11	764.548	21.257	54.2%	0.697	1349.641	41.671	5.118	44.4%	0.728	1263.031	8.110	2.359	46.7%	0.718	896.688
	Model 12	516.147	13.610	70.8%	0.799	993.344	36.851	4.853	50.0%	0.720	831.813	10.213	2.575	50.0%	0.666	693.766

Note: This table reports the error evaluation and time consumption of point prediction. MSE, MAE, SC and TIC are the evaluation indicators, which is shown in Eqs. 15, 16, 18 and 17. Models 1–12 represent PVMD-SVM-ARMA model, PVMD-SVM-ARMA-Bmax model, ARMA model, SVM model, EMD-SVM-ARMA model, EEMD-SVM-ARMA model, CEEMD-SVM-ARMA model, CEEMDAN-SVM-ARMA model, EMD-ELM-ARMA model, EEMD-ELM-ARMA model, CEEMD-ELM-ARMA model, CEEMDAN-ELM-ARMA model respectively.

Table 4
DM test results.

		Stage 1				Stage 2				Stage 3			
		MSE		MAE		MSE		MAE		MSE		MAE	
		Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
WTI	Model 1												
	Model 2	0.011		0.002		0.000		0.000		0.006		0.000	
	Model 3	0.004	0.024	0.000	0.055	0.003	0.464	0.000	0.600	0.037	0.039	0.000	0.001
	Model 4	0.001	0.024	0.000	0.050	0.009	0.403	0.000	0.459	0.041	0.043	0.001	0.002
	Model 5	0.001	0.029	0.000	0.058	0.008	0.687	0.004	0.793	0.034	0.036	0.001	0.002
	Model 6	0.004	0.068	0.001	0.112	0.006	0.405	0.000	0.564	0.032	0.035	0.000	0.001
	Model 7	0.021	0.110	0.001	0.120	0.000	0.159	0.000	0.321	0.031	0.033	0.000	0.000
	Model 8	0.005	0.071	0.002	0.127	0.001	0.503	0.000	0.517	0.034	0.036	0.000	0.001
	Model 9	0.001	0.022	0.000	0.048	0.001	0.490	0.000	0.461	0.034	0.036	0.001	0.002
	Model 10	0.000	0.037	0.000	0.036	0.012	0.206	0.000	0.279	0.033	0.035	0.001	0.002
	Model 11	0.000	0.009	0.000	0.003	0.000	0.000	0.000	0.000	0.004	0.005	0.000	0.000
	Model 12	0.000	0.034	0.000	0.040	0.028	0.068	0.002	0.102	0.034	0.036	0.001	0.002
Brent	Model 1												
	Model 2	0.010		0.001		0.005		0.005		0.001		0.000	
	Model 3	0.012	0.020	0.000	0.003	0.006	0.021	0.000	0.006	0.000	0.002	0.000	0.002
	Model 4	0.013	0.022	0.000	0.004	0.002	0.015	0.000	0.005	0.000	0.002	0.000	0.004
	Model 5	0.007	0.056	0.002	0.042	0.004	0.020	0.003	0.030	0.005	0.038	0.003	0.108
	Model 6	0.022	0.112	0.004	0.068	0.001	0.011	0.000	0.005	0.003	0.028	0.000	0.036
	Model 7	0.020	0.183	0.005	0.098	0.005	0.033	0.000	0.009	0.006	0.018	0.000	0.014
	Model 8	0.019	0.150	0.003	0.072	0.001	0.006	0.000	0.001	0.007	0.032	0.001	0.041
	Model 9	0.010	0.024	0.000	0.003	0.001	0.008	0.000	0.003	0.002	0.019	0.000	0.036
	Model 10	0.007	0.017	0.000	0.007	0.001	0.007	0.000	0.004	0.001	0.011	0.000	0.012
	Model 11	0.019	0.024	0.001	0.002	0.001	0.016	0.000	0.005	0.004	0.017	0.000	0.008
	Model 12	0.007	0.034	0.001	0.023	0.001	0.006	0.000	0.002	0.000	0.003	0.000	0.006
INE	Model 1												
	Model 2	0.004		0.000		0.003		0.000		0.004		0.001	
	Model 3	0.335	0.994	0.396	0.999	0.002	0.013	0.000	0.003	0.000	0.001	0.000	0.000
	Model 4	0.537	0.995	0.579	0.999	0.001	0.012	0.000	0.002	0.000	0.002	0.000	0.004
	Model 5	0.329	0.995	0.385	0.999	0.003	0.016	0.000	0.005	0.000	0.000	0.000	0.000
	Model 6	0.166	0.994	0.132	0.999	0.002	0.010	0.000	0.007	0.000	0.000	0.000	0.000
	Model 7	0.473	0.995	0.430	0.999	0.004	0.020	0.000	0.011	0.000	0.001	0.000	0.000
	Model 8	0.134	0.994	0.103	0.999	0.002	0.013	0.000	0.009	0.000	0.000	0.000	0.000
	Model 9	0.004	0.957	0.001	0.917	0.002	0.006	0.000	0.002	0.000	0.000	0.000	0.000
	Model 10	0.001	0.913	0.000	0.769	0.002	0.018	0.000	0.008	0.000	0.001	0.000	0.000
	Model 11	0.358	0.994	0.334	0.999	0.003	0.014	0.000	0.013	0.000	0.000	0.000	0.000
	Model 12	0.221	0.994	0.255	0.998	0.001	0.003	0.000	0.000	0.000	0.000	0.000	0.000

Note: This table reports the results of DM test. MSE and MAE are both the loss function of DM test. The null hypothesis of DM test is that the prediction efficiency of two models is consistent. Models 1–12 represent PVMD-SVM-ARMA model, PVMD-SVM-ARMA-Bmax model, ARMA model, SVM model, EMD-SVM-ARMA model, EEMD-SVM-ARMA model, CEEMD-SVM-ARMA model, CEEMDAN-SVM-ARMA model, EMD-ELM-ARMA model, EEMD-ELM-ARMA model, CEEMD-ELM-ARMA model, CEEMDAN-ELM-ARMA model respectively.

SVM-ARMA model are improved over 10% in almost all data sets. Excluding Stage 1 of INE, SCs of PVMD-SVM-ARMA model are all over 80% and more than that of Model 2–12. Besides, excluding INE Stage 1 and WTI Stage 2, PVMD-SVM-ARMA-Bmax model has an obvious advantage in MSE, MAE, SC, and TIC. For one thing, these findings from MSE, MAE, SC, and TIC indicate that PVMD-SVM-ARMA model is stable and reliable. For another, PVMD-SVM-ARMA-Bmax model has better prediction performance in most scenarios of different markets than other competing models. Finally, both PVMD-SVM-ARMA and PVMD-SVM-ARMA-Bmax models take a longer time in the prediction process with respect to the others. Long time consumption is normal and reasonable because PSO algorithm is used to optimize the VMD parameters in both models. Contrasting the prediction accuracy of PVMD-SVM-ARMA model and PVMD-SVM-ARMA-Bmax model, MSE, MAE, and TIC values of the latter model are all higher than these of the former model. Due to the uncertainties of noise information, the noise matrix will change the fluctuation range of predicted value, but it will not significantly improve the prediction accuracy, even make prediction accuracy decline. Therefore, the prediction results of PVMD-SVM-ARMA-Bmax model are reasonable and reliable.

When the confidence level is at 0.1, we take DM test and MCS test to contrast model performance. DM test results are reported in Table 4, and MCS test results are shown in Table 5–7. While PVMD-SVM-ARMA model is the benchmark model, Table 4 shows that all DM tests are

rejected except for some models in Stage 1 of INE. In the MCS test results, the rank of PVMD-SVM-ARMA model is always the first excluding Stage 1 of INE. Meanwhile, its p value is 1 all the time in the MCS tests of all data sets. All in all, it can be known that PVMD-SVM-ARMA model obviously outperforms other models in statistics from DM and MCS test results.

In different data sets, taking PVMD-SVM-ARMA-Bmax as the benchmark, we find that most of DM tests reject the null hypothesis. In MCS test, when there is more than one model in the optimal model set, PVMD-SVM-ARMA-Bmax model can often be selected. Combined with the error evaluation, DM test and MCS test, it could be considered that PVMD-SVM-ARMA-Bmax model is valid and credible. Therefore, the interval prediction results can be gained by combining PVMD-SVM-ARMA model and Bootstrap resampling, which not only quantifies noise information but also ensures interval prediction effectiveness.

Comparing the prediction effect of different models in different data sets, the single model consumption time is short, but the prediction error is generally larger (see Table 3). According to DM test and MCS test, hybrid models based on EMD, EEMD, CEEMD and CEEMDAN model occasionally perform well. For example, some models occasionally appear in the optimal model set of MCS test. However, the prediction error indicators of these models are far worse than these of PVMD-SVM-ARMA model. As can be seen from Table 3–7, PVMD-SVM-ARMA model greatly improves the prediction accuracy in the different frequencies. To

Table 5
MCS test results for WTI.

		TMAX				TR			
		MSE		MAE		MSE		MAE	
		rank	p value	rank	p value	rank	p value	rank	p value
Stage 1	Model 1	1	1.000	1	1.000	1	1.000	1	1.000
	Model 2	2	1.000	–	0.000	–	0.000	–	0.000
	Model 3	6	0.891	–	0.000	–	0.000	–	0.000
	Model 4	10	0.619	–	0.000	–	0.000	–	0.000
	Model 5	11	0.577	–	0.000	–	0.000	–	0.000
	Model 6	3	1.000	–	0.000	–	0.000	–	0.000
	Model 7	4	1.000	–	0.000	–	0.000	–	0.000
	Model 8	5	1.000	–	0.000	–	0.000	–	0.000
	Model 9	12	0.372	–	0.000	–	0.000	–	0.000
	Model 10	8	0.675	–	0.000	–	0.000	–	0.000
	Model 11	7	0.732	–	0.000	–	0.000	–	0.000
	Model 12	9	0.673	–	0.000	–	0.000	–	0.000
Stage 2	Model 1	1	1.000	1	1.000	1	1.000	1	1.000
	Model 2	6	1.000	5	1.000	–	0.000	–	0.000
	Model 3	5	1.000	3	1.000	–	0.000	–	0.000
	Model 4	7	1.000	8	0.996	–	0.000	–	0.000
	Model 5	2	1.000	2	1.000	–	0.000	–	0.000
	Model 6	4	1.000	4	1.000	–	0.000	–	0.000
	Model 7	10	1.000	9	0.983	–	0.000	–	0.000
	Model 8	3	1.000	6	1.000	–	0.000	–	0.000
	Model 9	8	1.000	7	0.999	–	0.000	–	0.000
	Model 10	9	1.000	10	0.700	–	0.000	–	0.000
	Model 11	–	–	–	–	–	0.000	–	0.000
	Model 12	11	0.375	11	0.522	–	0.000	–	0.000
Stage 3	Model 1	1	1.000	1	1.000	1	1.000	1	1.000
	Model 2	2	1.000	–	0.000	–	–	–	0.000
	Model 3	6	0.178	–	0.000	3	0.366	–	0.000
	Model 4	3	0.287	–	0.000	2	0.402	–	0.000
	Model 5	4	0.198	–	0.000	6	0.354	–	0.000
	Model 6	7	0.164	–	0.000	10	0.299	–	0.000
	Model 7	5	0.195	–	0.000	9	0.311	–	0.000
	Model 8	8	0.163	–	0.000	5	0.354	–	0.000
	Model 9	11	0.131	–	0.000	7	0.350	–	0.000
	Model 10	10	0.131	–	0.000	8	0.347	–	0.000
	Model 11	–	–	–	0.000	–	–	–	0.000
	Model 12	9	0.143	–	0.000	4	0.358	–	0.000

Note: This table reports MCS test results for WTI. MSE and MAE are both the loss functions of MCS test. In MCS test, the confidence level is 0.1, TMAX and TR are the statistics shown in Eqs. 20 and 21 respectively. The null hypothesis of MCS test is that these in the model set have equal predictive ability. Models 1–12 represent PVMD-SVM-ARMA model, PVMD-SVM-ARMA-Bmax model, ARMA model, SVM model, EMD-SVM-ARMA model, EEMD-SVM-ARMA model, CEEMD-SVM-ARMA model, CEEMDAN-SVM-ARMA model, EMD-ELM-ARMA model, EEMD-ELM-ARMA model, CEEMD-ELM-ARMA model, CEEMDAN-ELM-ARMA model respectively.

sum up, the proposed model has relatively reliable robustness with respect to the different frequency data.

An extreme event (COVID-19 pandemic) happened in 2020, which has a significant impact on the crude oil price (see [58]). Thus, the occurrence of COVID-19 might present challenges to predict crude oil price. We investigate the out-of-sample predictive performance during the business cycles to prove the effectiveness of proposed model, which is more attracted to both academic and industry community. More specifically, out-of-sample period is divided into two sub-periods, i.e., expansion and recession dated by the National Bureau of Economic Research⁴. For our forecast sets, the expansion period is from July 2009 to January 2020 and recession period starts from February 2020, in which COVID-19 period is the same as the recession. In the meantime, out-of-sample prediction of WTI and Brent are included in these two periods, and these of INE are all located in the COVID-19 period. Loss evaluation during these two sub-periods for WTI, Brent, and INE is made

⁴ <https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions>.

and shown in Table 8⁵. For one thing, relative to other hybrid models, PVMD-SVM-ARMA model has better prediction performance and its SCs are all over 80% in the COVID-19 period. Therefore, PVMD-SVM-ARMA model also performs well even in the COVID-19, which suggests that our model is stable even though facing the global shock. For another thing, PVMD-SVM-ARMA model has good prediction accuracy in both periods. Therefore, it can be considered that PVMD-SVM-ARMA model has good robustness in expansion and recession periods. In summary, even if global event such as COVID-19 occurs, the proposed model is also valid to some extent.

VMD has more advantages in denoising and can decompose the data into more regular sequences, which might be the reason why the hybrid model based on VMD outperforms other models. The hybrid model based on ARMA model and SVM model is contrasted with the hybrid model based on ARMA model and ELM model. The former performs better than the latter generally in these experiments. The possible reason is that, with ELM model compared, SVM model has obvious advantages in small sample prediction. Contrasting with the single model, hybrid

⁵ There are the drastic fluctuation from March to May, 2020. Models 3–12 are not stable, and uncertain events will have a great impact on the prediction accuracy of these models. Therefore, models 3–12 produce large error in the recession of WTI.

Table 6
MCS test results for Brent.

		TMAX				TR			
		MSE		MAE		MSE		MAE	
		rank	p value	rank	p value	rank	p value	rank	p value
Stage 1	Model 1	1	1.000	1	1.000	1	1.000	1	1.000
	Model 2	2	1.000	–	0.000	–	0.000	–	0.000
	Model 3	8	0.120	–	0.000	–	0.000	–	0.000
	Model 4	7	0.137	–	0.000	–	0.000	–	0.000
	Model 5	5	1.000	–	0.000	–	0.000	–	0.000
	Model 6	6	1.000	–	0.000	–	0.000	–	0.000
	Model 7	3	1.000	–	0.000	–	0.000	–	0.000
	Model 8	4	1.000	–	0.000	–	0.000	–	0.000
	Model 9	–	–	–	0.000	–	0.000	–	0.000
	Model 10	–	–	–	0.000	–	0.000	–	0.000
	Model 11	–	–	–	0.000	–	0.000	–	0.000
	Model 12	9	0.119	–	0.000	–	0.000	–	0.000
Stage 2	Model 1	1	1.000	1	1.000	1	1.000	1	1.000
	Model 2	–	0.000	–	0.000	–	0.000	–	0.000
	Model 3	–	0.000	–	0.000	–	0.000	–	0.000
	Model 4	–	0.000	–	0.000	–	0.000	–	0.000
	Model 5	–	0.000	–	0.000	–	0.000	–	0.000
	Model 6	–	0.000	–	0.000	–	0.000	–	0.000
	Model 7	–	0.000	–	0.000	–	0.000	–	0.000
	Model 8	–	0.000	–	0.000	–	0.000	–	0.000
	Model 9	–	0.000	–	0.000	–	0.000	–	0.000
	Model 10	–	0.000	–	0.000	–	0.000	–	0.000
	Model 11	–	0.000	–	0.000	–	0.000	–	0.000
	Model 12	–	0.000	–	0.000	–	0.000	–	0.000
Stage 3	Model 1	1	1.000	1	1.000	1	1.000	1	1.000
	Model 2	2	1.000	–	0.000	–	0.000	–	0.000
	Model 3	9	0.343	–	0.000	–	0.000	–	0.000
	Model 4	5	0.991	–	0.000	–	0.000	–	0.000
	Model 5	3	1.000	–	0.000	–	0.000	–	0.000
	Model 6	4	1.000	–	0.000	–	0.000	–	0.000
	Model 7	10	0.316	–	0.000	–	0.000	–	0.000
	Model 8	6	0.978	–	0.000	–	0.000	–	0.000
	Model 9	8	0.869	–	0.000	–	0.000	–	0.000
	Model 10	7	0.975	–	0.000	–	0.000	–	0.000
	Model 11	11	0.215	–	0.000	–	0.000	–	0.000
	Model 12	12	0.183	–	0.000	–	0.000	–	0.000

Note: This table reports MCS test results for Brent. MSE and MAE are both the loss functions of MCS test. In MCS test, the confidence level is 0.1, TMAX and TR are the statistics shown in Eqs. 20 and 21 respectively. The null hypothesis of MCS test is that these in the model set have equal predictive ability. Models 1–12 represent PVMD-SVM-ARMA model, PVMD-SVM-ARMA-Bmax model, ARMA model, SVM model, EMD-SVM-ARMA model, EEMD-SVM-ARMA model, CEEMD-SVM-ARMA model, CEEMDAN-SVM-ARMA model, EMD-ELM-ARMA model, EEMD-ELM-ARMA model, CEEMD-ELM-ARMA model, CEEMDAN-ELM-ARMA model respectively.

model can describe simultaneously the multiple characteristics such as nonlinearity and linearity. In conclusion, it is reliable that PVMD-SVM-ARMA model with kinds of advantages has good prediction accuracy and robustness.

4.2.3. The analysis of point prediction

Fig. 3 displays the online real-time prediction results and errors, in which the left three subgraphs of (a), (b) and (c) are online point prediction results in three stages, and these on the right correspond to the prediction errors from Stage 1 to Stage 3. In the left three subgraphs, the black line and red line are the actual and prediction curve respectively, and the blue rectangle marked with numbers is the point with greater prediction errors. In WTI data set, the absolute error values of marked points are more than 8. In Brent data set, the absolute error values greater than 5 are taken as the standard to select the marked points. In INE data set, the points with absolute error over 2 are marked.

In the fluctuation process, there will be a peak after the dramatic rise and a trough after a sharp fall. For WTI forecast, there are absolute error values of five points over 8 in Stage 1, and they are located at the end of July 2015, January, February, April, and July 2016 respectively. In Stage 1, point 1, point 4, and point 5 are in the trough or peak, and points 2, 3 are near the peak. The point with absolute error over 8 is seated on November 30, 2018 for Stage 2, and these points in Stage 3 are situated at the late of March, May, August, September and October 2020.

Combining with data of Stage 3, point 1 of the second stage occurs in the trough. Meantime, the first two maximum errors of Stage 3, namely points 1 and 2, are located in the trough and the peak, and the remaining points are located near the peak. For Brent prediction, in Stage 1, there are larger errors on February 28, March 6, 13, and 20, 2020, and the first three maximum errors of Stage 2 are on April 3, June 12, and July 24, 2020. Uniting with the curve of Stage 2, it can be known that, point 3 of Stage 1 is in the trough, and others are near the trough. Besides, point 1 in Stage 2 is located at the peak. For INE forecast, the marked point is that the absolute error is over 2, which is located on July 21, August 25, October 16 and November 25, 2020. Compared with monthly and weekly data, daily data is high-frequency data. When uncontrollable global events occur, the daily data will fluctuate up and down, falling or rising continuously, which causes the peaks and troughs occurring in the long-term rather than in the short-term more obviously. Therefore, in INE, the points with large errors do not present obvious position characteristics. However, the greater error points in Stage 1 and 2 of INE are consistent with point 3 in Stage 2 of Brent and points 3, 5 in Stage 3 of WTI. In conclusion, the points with greater forecasting errors are mainly concentrated in the peaks and troughs or somewhere around them, and the principal reason might be that uncontrollable global events will increase the prediction difficulty.

The undesirable forecasting results during the peaks and troughs might be related to the impacts of extreme events. Especially, in Stage 1

Table 7
MCS test results for INE.

		TMAX				TR			
		MSE		MAE		MSE		MAE	
		rank	p value	rank	p value	rank	p value	rank	p value
Stage 1	Model 1	3	1.000	5	1.000	2	1.000	2	1.000
	Model 2	–	–	–	–	–	–	–	–
	Model 3	6	1.000	3	1.000	7	0.475	7	0.754
	Model 4	1	1.000	1	1.000	1	1.000	1	1.000
	Model 5	5	1.000	2	1.000	5	0.823	3	0.938
	Model 6	8	0.398	8	0.232	9	0.281	8	0.227
	Model 7	2	1.000	4	1.000	3	0.997	6	0.790
	Model 8	9	0.289	9	0.225	8	0.289	–	–
	Model 9	–	–	–	–	–	–	–	–
	Model 10	–	–	–	–	–	–	–	–
	Model 11	4	1.000	6	1.000	4	0.965	5	0.853
	Model 12	7	0.818	7	0.985	6	0.808	4	0.917
Stage 2	Model 1	1	1.000	1	1.000	1	1.000	1	1.000
	Model 2	2	1.000	2	1.000	–	0.000	–	0.000
	Model 3	6	0.274	8	0.832	–	0.000	–	0.000
	Model 4	7	0.245	11	0.397	–	0.000	–	0.000
	Model 5	4	0.523	6	0.964	–	0.000	–	0.000
	Model 6	5	0.340	5	0.979	–	0.000	–	0.000
	Model 7	10	0.178	7	0.856	–	0.000	–	0.000
	Model 8	8	0.217	3	0.995	–	0.000	–	0.000
	Model 9	9	0.201	10	0.420	–	0.000	–	0.000
	Model 10	3	0.737	4	0.995	–	0.000	–	0.000
	Model 11	–	–	9	0.693	–	0.000	–	0.000
	Model 12	–	–	12	0.114	–	0.000	–	0.000
Stage 3	Model 1	1	1.000	1	1.000	1	1.000	1	1.000
	Model 2	–	0.000	–	0.000	–	0.000	–	0.000
	Model 3	–	0.000	–	0.000	–	0.000	–	0.000
	Model 4	–	0.000	–	0.000	–	0.000	–	0.000
	Model 5	–	0.000	–	0.000	–	0.000	–	0.000
	Model 6	–	0.000	–	0.000	–	0.000	–	0.000
	Model 7	–	0.000	–	0.000	–	0.000	–	0.000
	Model 8	–	0.000	–	0.000	–	0.000	–	0.000
	Model 9	–	0.000	–	0.000	–	0.000	–	0.000
	Model 10	–	0.000	–	0.000	–	0.000	–	0.000
	Model 11	–	0.000	–	0.000	–	0.000	–	0.000
	Model 12	–	0.000	–	0.000	–	0.000	–	0.000

Note: This table reports MCS test results for INE. MSE and MAE are both the loss functions of MCS test. In MCS test, the confidence level is 0.1, TMAX and TR are the statistics shown in Eqs. 20 and 21 respectively. The null hypothesis of MCS test is that these in the model set have equal predictive ability. Models 1–12 represent PVMD-SVM-ARMA model, PVMD-SVM-ARMA-Bmax model, ARMA model, SVM model, EMD-SVM-ARMA model, EEMD-SVM-ARMA model, CEEMD-SVM-ARMA model, CEEMDAN-SVM-ARMA model, EMD-ELM-ARMA model, EEMD-ELM-ARMA model, CEEMD-ELM-ARMA model, CEEMDAN-ELM-ARMA model respectively.

of WTI, three black swan events including the escalation of Greek debt crisis, China's stock market crash, and Iran nuclear deal broke out in July 2015, which led to a sharp drop in crude oil price, and the predicted value of point 1 in Stage 1 explains these event impact. In mid-February 2016, four major oil producers including Saudi Arabia and Russia reached a consensus in freezing oil production, which resulted in oil price rise. In April and May 2016, strikes by oil workers in Nigeria and political unrest in Venezuela affected oil supplies, which also contributed to the rise of crude oil price. In Fig. 3 (a), the predicted log-return over 0 from the late of February to May 2016 reflects the above price increase. In November 2018, U.S. sanctions against Iran's oil industry went into effect, and the U.S. president Trump put pressure on the Organization of Petroleum Exporting Countries (OPEC) on Twitter to not support oil prices by reducing supply, which caused a new round of selling and led to a sharp drop in oil prices. The prediction value of point 1 in Stage 2 of Fig. 3 (a) is far less than 0, which illustrates this event. From the February to the end of March 2020, the COVID-19 began to break out around the world. The global demand for oil products decreased, and the oil price dropped sharply in the first quarter of 2020. The impact of this event is reflected by the prediction value of point 1 in Stage 3 of Fig. 3 (a). Besides, the forecasting values of points 1, 2, 3, and 4 in Stage 1 of Fig. 3 (b) are all less than 0, which also embodies the influence. In May 2020, the OPEC + production cut agreement has come into effect, and the decline in crude oil supply has resulted in a

significant increase in crude oil price. The prediction value of point 2 in WTI Stage 3 and the predicted log-return over 0 of Brent in May explain the fact. The occurrence of uncertain global events will bring challenges to the prediction task and decline the forecasting accuracy. However, combined with Table 8 and Fig. 3, it can be known that the proposed method can well describe the fluctuations in crude oil prices even if uncertain events (including the COVID-19) occur.

When uncontrollable global events have a serious impact on crude oil price, structural break could occur. Since the emergence of COVID-19 in January 2020, international oil prices have been plummeted and it keep rising subsequently (as shown in Fig. 2), which has the characteristic of structural break. And Atukeren et al. [59] testified that the structural breaks of WTI and Brent occurred in March and June respectively. Therefore, referring to [34], we also make predictions when the structural break occurs in 2020, which can test the validity of proposed model. Structural breaks will take a great challenge for prediction. Nevertheless, according to Fig. 3 and SC of Table 3, PVMD-SVM-ARMA model can predict the positive and negative of crude oil log-return well. That is, our forecasting method can be employed to judge the rise and fall of crude oil price even in the structural break.

4.2.4. The analysis of interval prediction

The interval prediction results from Stage 1 to Stage 3 are shown in Fig. 4. In the interval prediction, the trend of some points is not in line

Table 8

The evaluation of point prediction in expansion and recession periods.

		Expansion				Recession (COVID-19)			
		MSE	MAE	SC	TIC	MSE	MAE	SC	TIC
WTI	Model 1	18.147	3.129	85.5%	0.254	85.302	7.347	90.0%	0.150
	Model 2	53.079	5.400	74.2%	0.427	120.893	9.009	90.0%	0.182
	Model 3	96.965	7.446	54.8%	0.810	1130.975	22.386	70.0%	0.909
	Model 4	98.760	7.671	54.8%	0.919	1154.935	22.373	60.0%	0.960
	Model 5	88.071	7.125	59.7%	0.680	1068.738	21.580	50.0%	0.838
	Model 6	89.131	7.330	54.8%	0.666	1082.254	21.876	50.0%	0.836
	Model 7	96.842	7.703	59.7%	0.633	1130.060	23.809	50.0%	0.772
	Model 8	86.543	7.313	54.8%	0.670	1081.274	21.725	50.0%	0.854
	Model 9	97.843	7.672	58.1%	0.669	1079.314	21.634	70.0%	0.870
	Model 10	100.772	7.748	56.5%	0.707	1107.370	22.082	50.0%	0.870
	Model 11	303.189	14.068	59.7%	0.705	1403.477	29.094	60.0%	0.737
	Model 12	151.208	8.661	58.1%	0.626	1097.019	21.916	60.0%	0.877
Brent	Model 1	3.085	1.271	81.8%	0.219	16.386	2.827	93.0%	0.268
	Model 2	5.664	1.792	81.8%	0.309	30.722	4.098	81.4%	0.381
	Model 3	17.101	3.276	81.8%	0.934	83.927	7.168	41.9%	0.837
	Model 4	17.501	3.308	54.5%	0.949	78.484	6.929	51.2%	0.981
	Model 5	20.165	3.453	72.7%	0.703	62.888	5.873	62.8%	0.637
	Model 6	8.005	2.409	72.7%	0.405	66.920	6.489	55.8%	0.668
	Model 7	8.410	2.546	72.7%	0.446	66.927	6.500	53.5%	0.632
	Model 8	9.202	2.663	72.7%	0.460	66.333	6.483	60.5%	0.663
	Model 9	47.364	5.225	63.6%	0.813	86.052	7.147	48.8%	0.646
	Model 10	20.208	3.503	72.7%	0.719	68.674	6.626	58.1%	0.777
	Model 11	23.214	4.057	63.6%	0.460	134.090	8.354	46.5%	0.721
	Model 12	18.976	3.350	72.7%	0.699	69.395	6.739	53.5%	0.720
INE	Model 1	–	–	–	–	1.120	0.834	80.0%	0.252
	Model 2	–	–	–	–	6.133	1.678	62.2%	0.496
	Model 3	–	–	–	–	6.078	1.791	38.9%	0.932
	Model 4	–	–	–	–	5.770	1.732	55.6%	0.895
	Model 5	–	–	–	–	5.074	1.677	56.7%	0.613
	Model 6	–	–	–	–	5.302	1.721	53.3%	0.630
	Model 7	–	–	–	–	5.387	1.717	58.9%	0.693
	Model 8	–	–	–	–	5.612	1.742	51.1%	0.653
	Model 9	–	–	–	–	8.217	2.222	61.1%	0.640
	Model 10	–	–	–	–	6.640	2.063	51.1%	0.622
	Model 11	–	–	–	–	5.825	1.752	56.7%	0.710
	Model 12	–	–	–	–	7.878	2.059	53.3%	0.672

Note: This table reports the loss functions of Models 1–12 in the expansion and recession periods. MSE, MAE, SC and TIC are the evaluation indicators, which is shown in Eqs. 15, 16, 18 and 17. Models 1–12 represent PVMD-SVM-ARMA model, PVMD-SVM-ARMA-Bmax1model, ARMA model, SVM model, EMD-SVM-ARMA model, EEMD-SVM-ARMA model, CEEMD-SVM-ARMA model, CEEMDAN-SVM-ARMA model, EMD-ELM-ARMA model, EEMD-ELM-ARMA model, CEEMD-ELM-ARMA model, CEEMDAN-ELM-ARMA model respectively.

with that of original sequence, which is marked by green rectangles. The uncovered original points in the interval are circled in red. Table 9 is the interval prediction evaluation of three stages. The coverage rates of actual value in the three stages are all more than 70%, the interval diameters of WTI are all less than 20, and none of these in Brent are more than 11. In Stage 1 of INE, the interval diameters D and the fluctuation consistency C do not perform very well, but these values are also in the acceptable range. In other word, the interval prediction results are reasonable and reliable.

As can be seen from Fig. 4, for WTI, the uncovered points 1, 2, and 5 in Stage 1 concentrate in the troughs, the uncovered points 3 and 4 locate near the peak. The uncovered point 1 in Stage 2 is in the trough. In Stage 3, the uncovered point 3 and 4 gather in the trough and peak. For Brent, points 1, 2, 3, and 4 in Stage 1 cluster around the trough, and point 1 of Stage 2 is in the peak. For Stage 3 of INE, there is not evident peak or trough. In brief, the uncovered points mainly get together in or near the peaks and troughs. The peaks and troughs are sensitive and unstable, which means that their information is complex and unpredictable. Thereupon it's more difficult for us to make interval prediction in the points with greater fluctuation.

Although this study cannot completely cover all the points of test set, the fluctuation trend of other uncovered points, except points 5 and 7 in Stage 3 of WTI, are all in line with the actual value. The interval prediction results can identify the fluctuation characteristics in the peaks and troughs. In addition, most of the fluctuation consistency indicators

are more than 75%, in Table 9, which demonstrates that PVMD-SVM-ARMA-B model has advantages in describing the series fluctuation characteristics. In a word, the interval prediction results are highly reliable, and can provide a scientific basis for investors' decision-making in actual investment.

According to Section 4.2.2, the out-of-sample are divided into expansion and recession periods. Table 10⁶ shows the interval forecast evaluation in expansion and recession periods. In the expansion period, the interval coverage rate and the fluctuation consistency are both more than 80%. In the recession (COVID-19) period, the interval coverage of WTI decreases significantly, but all fluctuation consistency indicators of three data sets are more than 70%. Comparing the r_{cov} and C of two periods, PVMD-SVM-ARMA-B performs better in the expansion period. The possible reason is that COVID-19 weakens the prediction performance of PVMD-SVM-ARMA-B model. However, in the COVID-19 period, the interval coverage in Brent and INE are both over 80%, and both fluctuation consistency is greater than 70%. Although the interval

⁶ When the out-of-sample is divided into expansion and recession periods, the reference value changes when calculating C_l and C_u . Therefore, in the three data sets, the fluctuation consistency of first value in Stage 2 and 3 is taken into account. When the upper bound and lower bound of these values fluctuate inconsistently, C_l and C_u change. It is why there are inconsistencies between C_l and C_u in Tables 9 and 10.

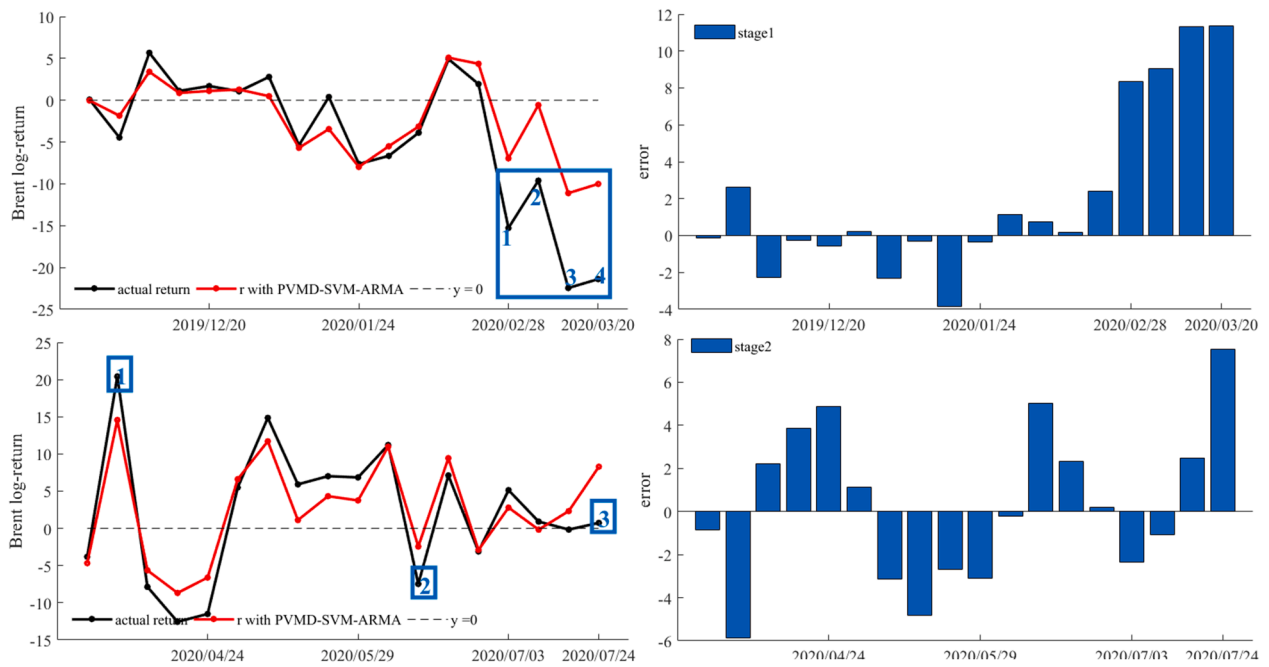
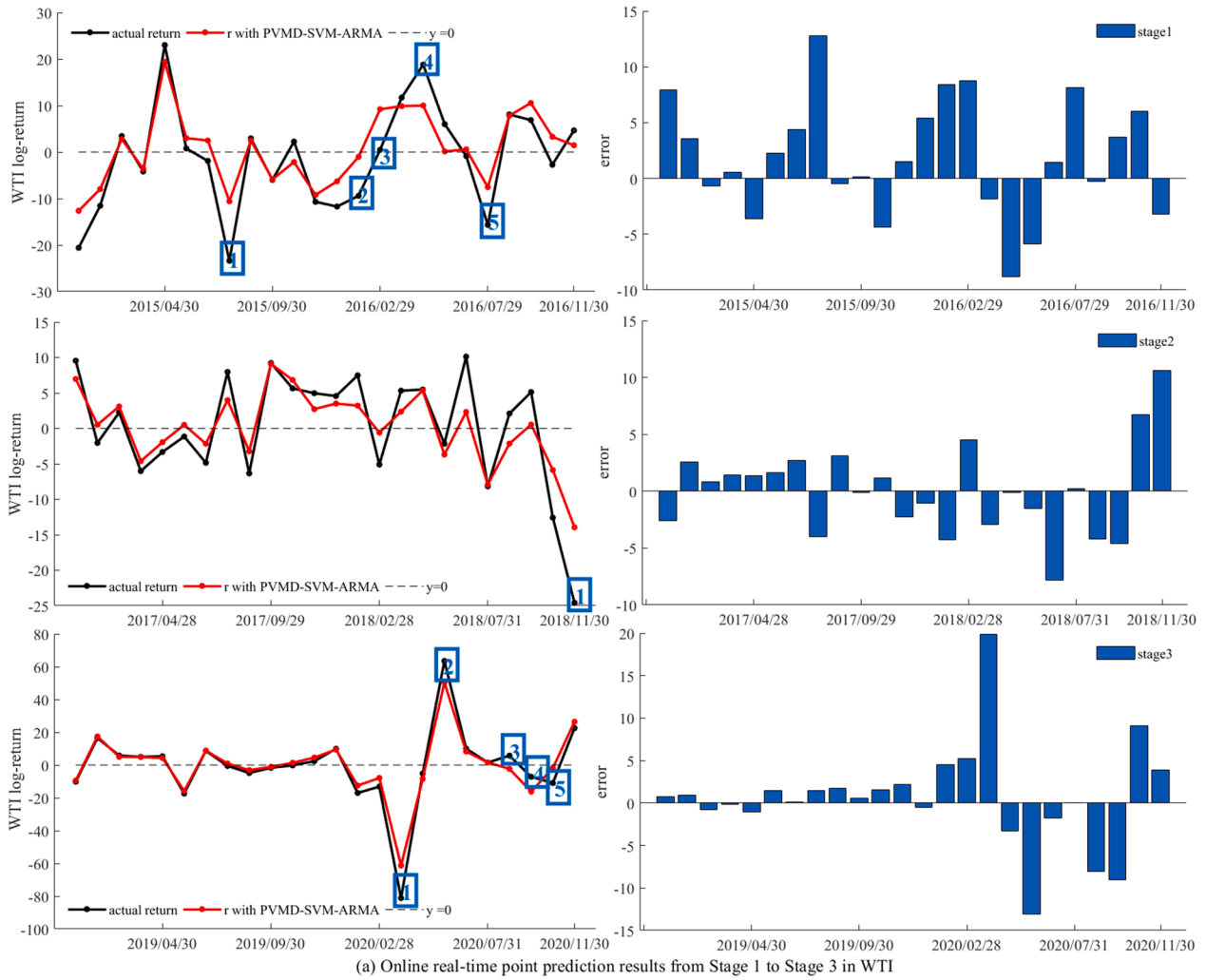


Fig. 3. Online real-time point prediction results and errors.

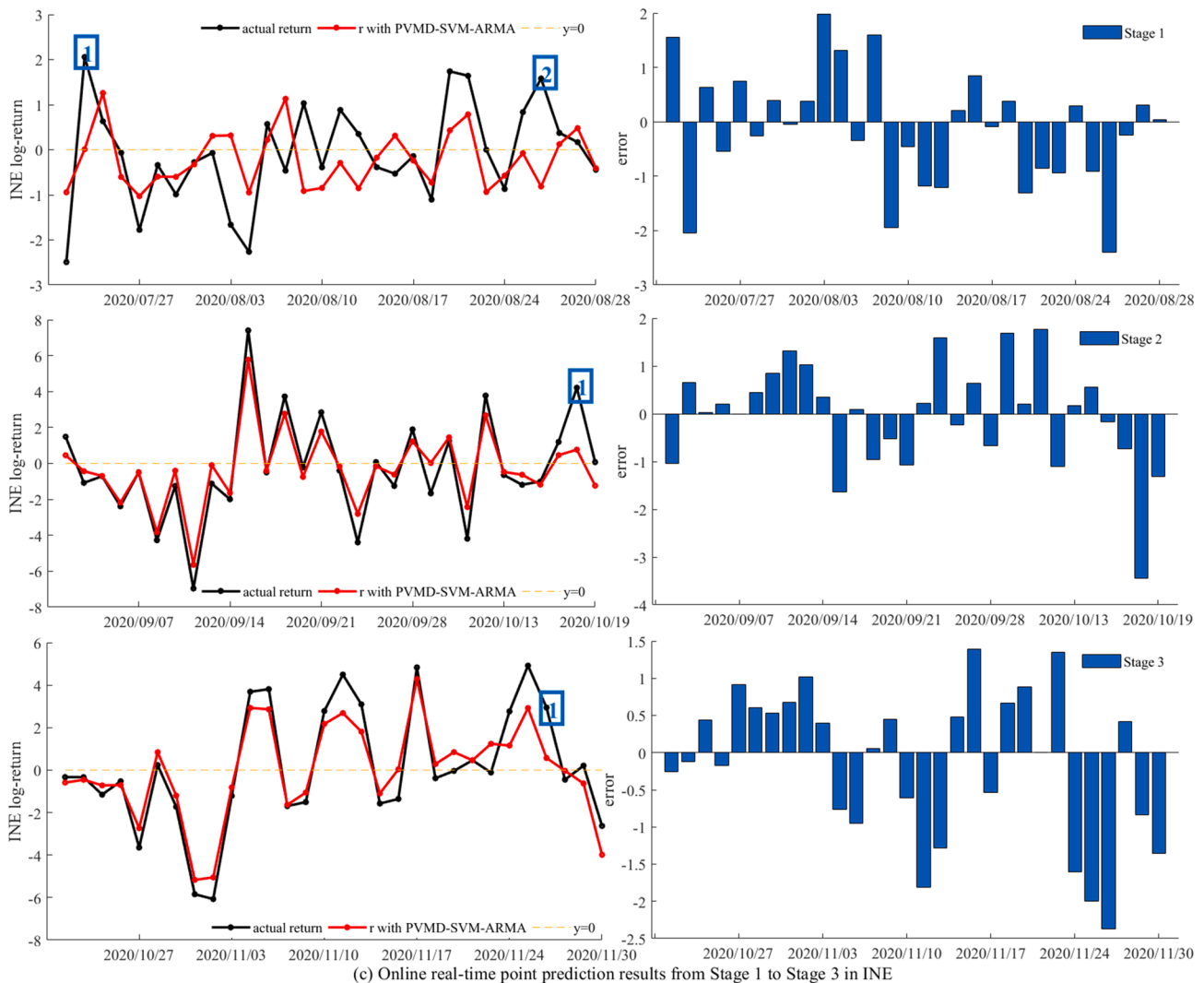
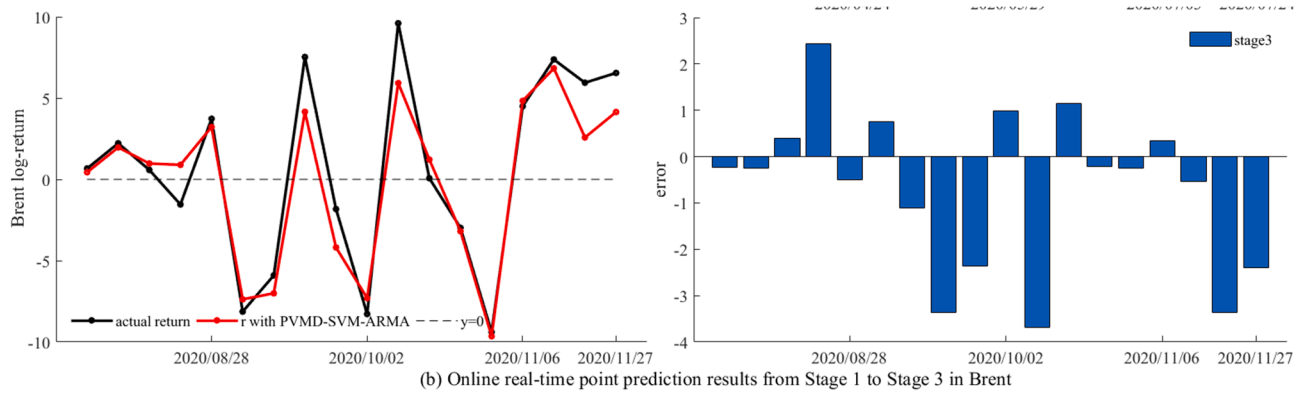


Fig. 3. (continued).

coverage indicator of WTI is not very good, the fluctuation consistency indicators exceed 75%, and the fluctuation trend of log-return is also well judged according to the interval prediction. In conclusion, PVMD-SVM-ARMA-B has good robustness and can well describe the fluctuation of log-return even if facing the global shock such as COVID-19.

Fluctuation consistency C at the interval boundary is the similar in Table 9 and 10. There are two reasons. The first one is that the noise matrix obtained by Bootstrap resampling will make the boundary tend to be stable. By comparing Fig. 9 and Fig. 10, it can be found that the

greater error of point prediction almost include all the uncovered points in interval prediction. Thus, another one is that the interval prediction results of PVMD-SVM-ARMA-B model depend on the point prediction of PVMD-SVM-ARMA model. Thus, there is a uniform fluctuation in the interval boundary formed by the maximum and minimum values.

Based on the PVMD-SVM-ARMA model, log-return of crude oil can be predicted well. The results show that the proposed model for point prediction has prominent prediction performance. Thanks to the uncertain characteristics of noise, it is difficult to capture and predict.

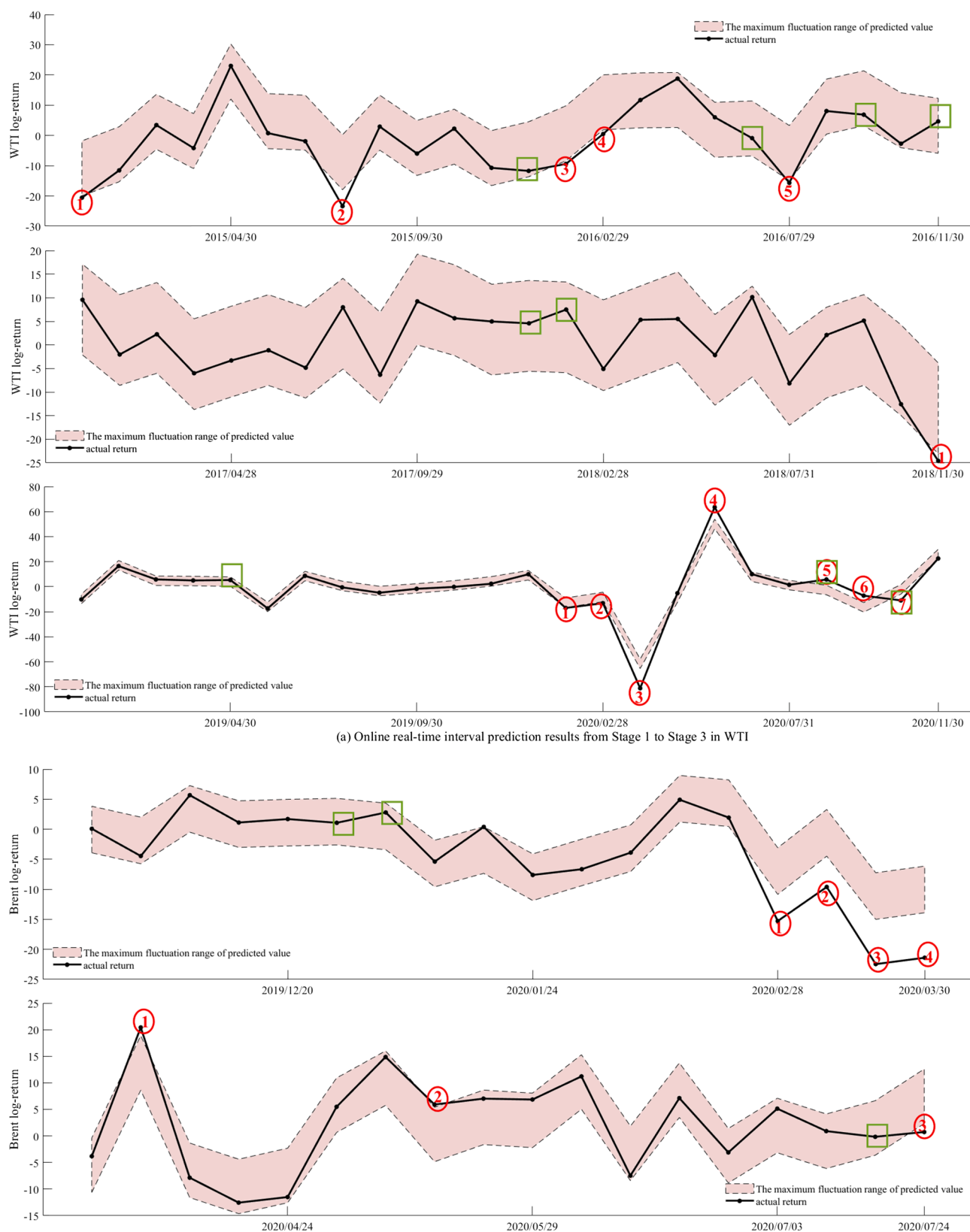


Fig. 4. The interval prediction result from Stage 1 to Stag 3.

Utilizing Bootstrap resampling and interval prediction, the noise information can be quantified, and the results are credible and can identify the fluctuation tendency of sequence partly.

4.2.5. The application of crude oil forecasting

As the main energy and basic resources of petrochemical industry, crude oil is not only inextricably linked with other energy and chemical

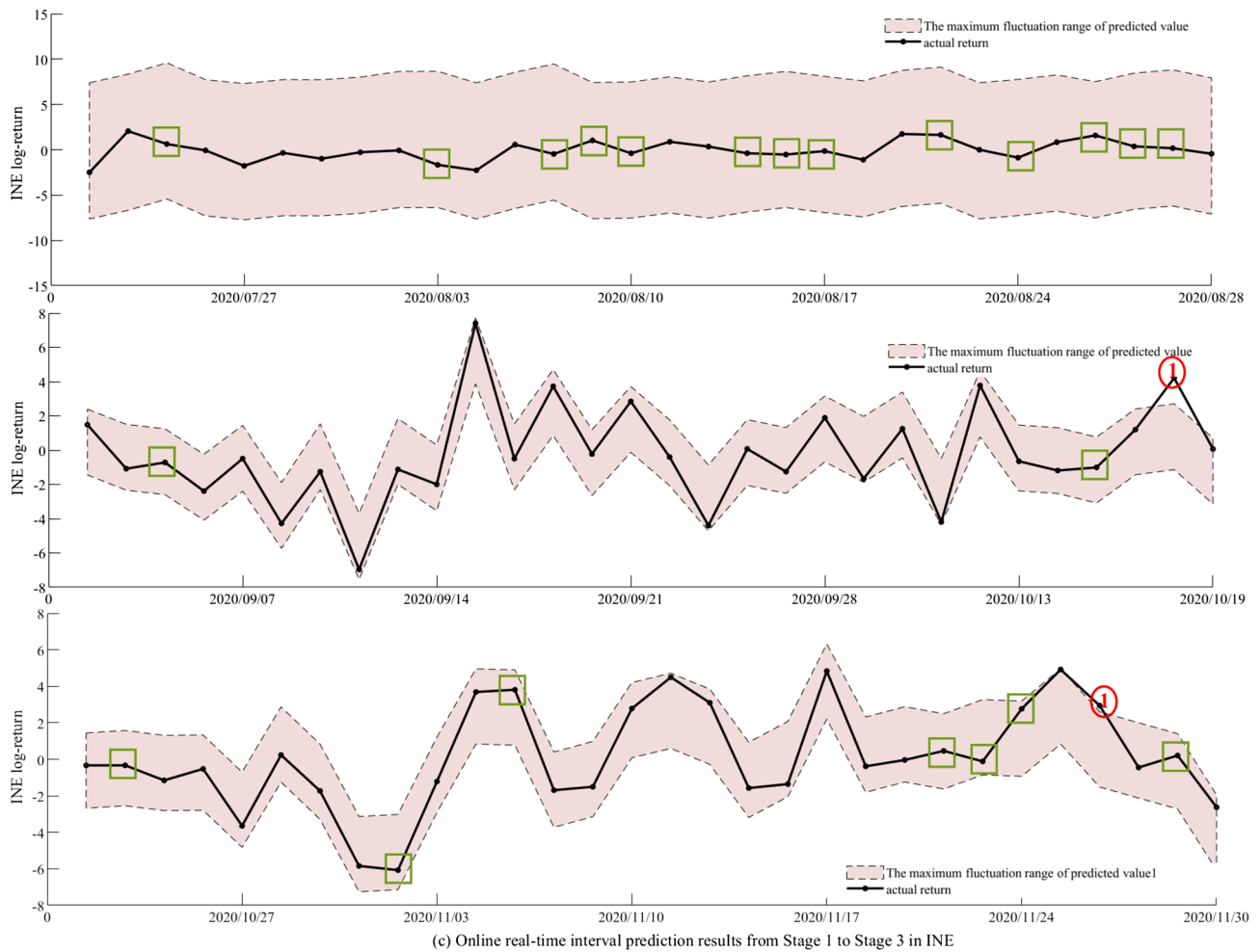
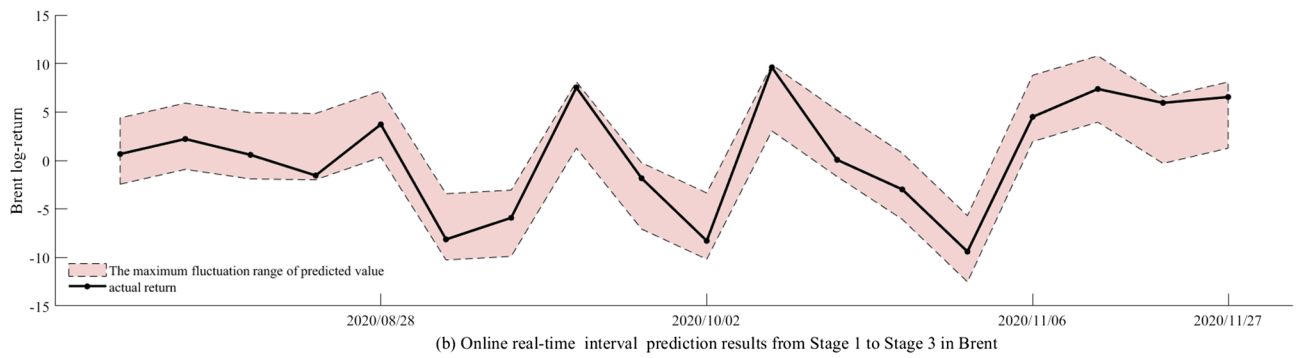


Fig. 4. (continued).

Table 9

The interval prediction evaluation of three stages.

Asset	D			r_{cov}			C_l			C_u		
	Stage 1	Stage 2	Stage 3	Stage 1	Stage 2	Stage 3	Stage 1	Stage 2	Stage 3	Stage 1	Stage 2	Stage 3
WTI	18.173	19.232	7.545	79.2%	95.8%	70.8%	82.6%	91.3%	87.0%	82.6%	91.3%	87.0%
Brent	7.754	10.256	6.832	77.8%	83.3%	100.0%	88.2%	94.1%	100.0%	88.2%	94.1%	100.0%
INE	15.018	3.841	4.121	100.0%	96.7%	96.7%	55.2%	93.1%	75.9%	55.2%	93.1%	75.9%

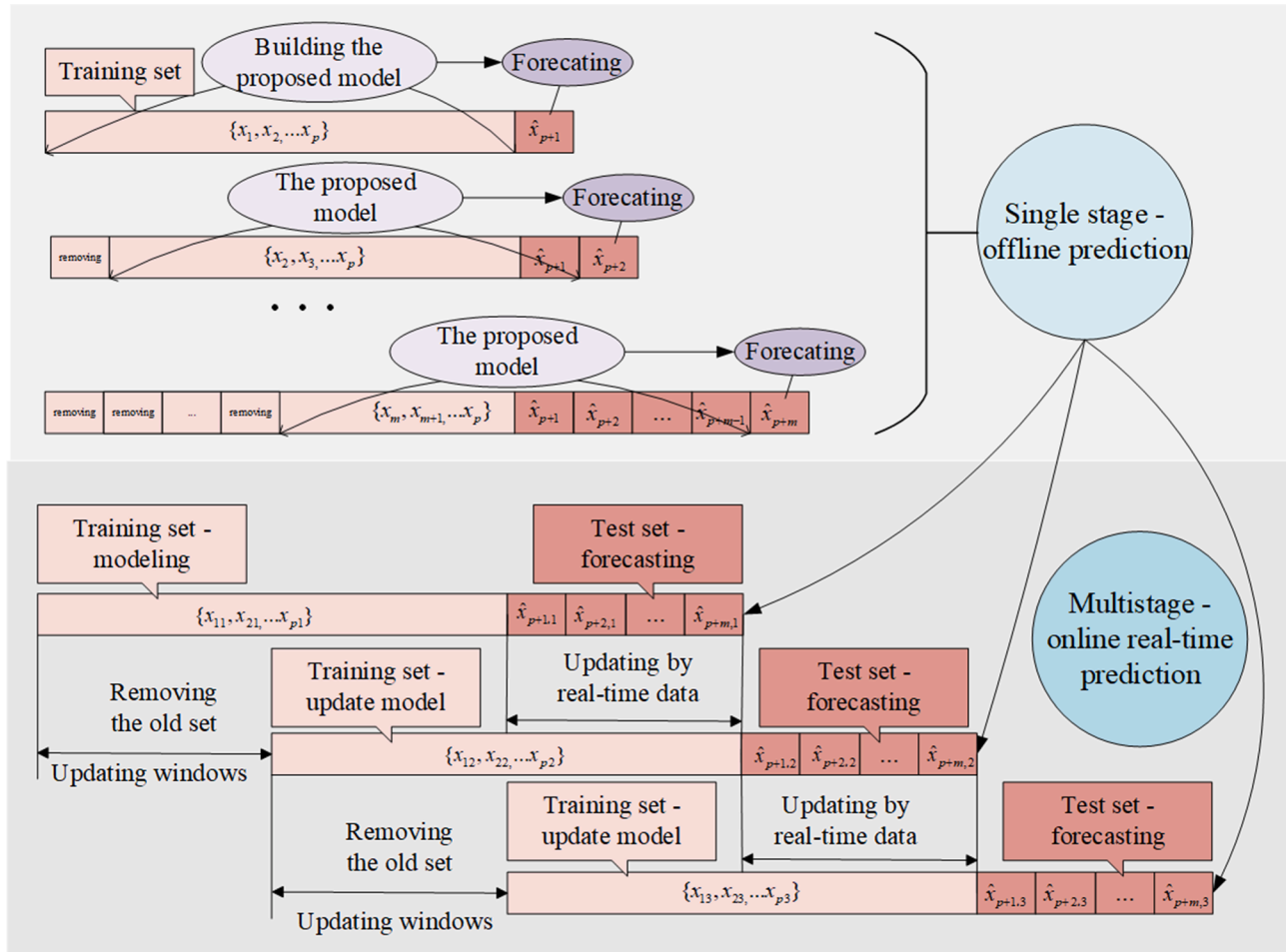
Note: This table reports the interval prediction evaluation of three stages. D , r_{cov} , C_l and C_u are the evaluation indicators. D , r_{cov} , C_l and C_u are the evaluation indicators, which is shown in Eq. 23, 22 and 24. According to Eq. 24, the first value of each stage is the reference value, so its fluctuation consistency is not calculated.

Table 10

The evaluation of interval prediction in expansion and recession.

	Expansion				-	Recession (COVID-19)			
	D	r_{cov}	C_l	C_u		D	r_{cov}	C_l	C_u
WTI	16.183	88.7%	88.5%	86.9%		7.545	40.0%	77.8%	77.8%
Brent	7.754	100.0%	80.0%	80.0%		8.415	83.7%	97.6%	97.6%
INE	—	—	—	—		7.660	97.8%	74.2%	73.0%

Note: This table reports the evaluation of interval prediction in expansion and recession. D , r_{cov} , C_l and C_u are the evaluation indicators, which is shown in Eqs. 23, 22 and 24.

**Fig. C.5.** Offline and online forecasting scheme.

products, but also affects the development of global trade. Accurate and reliable forecast of crude oil price is of great practical significance. First of all, crude oil is one of the energy suppliers, and various industrial departments cannot work without crude oil products. Therefore, crude oil price forecast provides a reference for industrial manufacturing to adjust relevant budgets. Secondly, crude oil price will affect the related energy products. For example, natural gas demand is closely related crude oil price. The sharp decline in international oil prices will weaken the price advantage and economy of natural gas, which will restrain the growth rate of natural gas demand. Thirdly, crude oil, as a kind of investment product, accurate price prediction helps investors avoid risk and get profit. In summary, crude oil price forecasting plays a significant role in energy product market and energy investment.

The proposed method in this paper can well predict the trend of crude oil prices in the US, UK and China. From the global perspective, WTI and Brent, as the international crude oil, affect the oil markets of other countries. For one thing, international crude oil forecasting can

help the traders who are related to global crude oil materials to understand its price tendency, and it can provide a basis for them to adjust their financial budget. For another thing, crude oil as an international investment product, accurate and reliable crude oil price forecast can help investors to judge the investment opportunities of crude oil and related financial products such as golden. From the perspective of China, INE comes from Chinese crude oil market, which is related to the international crude oil market, but there are also differences. For investors in Chinese oil market, they are more concerned about domestic oil prices, and INE crude oil price provides a more direct reference than WTI or Brent. For Chinese government, high prediction performance of INE crude oil price is in favour of industrial restructuring and policy regulation. To sum up, we can judge the trend of crude oil prices in different countries according to our forecasting results, which provides decision basis for both international and Chinese crude oil watchers.

5. Conclusion

Focusing on improving prediction accuracy, quantifying noise information and accomplishing online real-time prediction, this paper proposes a new crude oil price forecasting method. Firstly, the optimized variation mode decomposition is used to decompose the crude oil price time series, and then the sub-models are selected according to the data characteristics, and they are reconstructed to form the proposed point prediction model (PVMD-SVM-ARMA). Meanwhile, interval prediction model (PVMD-SVM-ARMA-B) is got by combining with PVMD-SVM-ARMA model and Bootstrap sampling. Finally, the hybrid model is formed into a predictor, so as to realize the model updating with data changing.

Different frequency data are used for empirical analysis. The empirical analysis can draw the following conclusions. (1) PVMD-SVM-ARMA model is superior to other models in both different frequency data and different periods (expansion and recession period). That is, it has better prediction accuracy and robustness. (2) PVMD-SVM-ARMA model can well predict the trend of crude oil prices even if structural breaks or global shock occur. (3) PVMD-SVM-ARMA-B model can be used for the interval prediction of crude oil price, which quantifies the uncertainty of noise information. (4) According to the proposed forecasting method, it can provide reference information for crude oil followers to make favorable decisions, such as those who work with crude oil and investor in energy financial products.

6. Future work

Although the prediction method in this paper has some advantages, there are still some interesting questions waiting to be explored, which can be concluded as follows. (1) The method proposed in this paper is based on the ARMA model and SVM model, so the method aims at small samples and short-term predictions mainly. However, whether the presented models perform well in the large samples and long-term predictions still leaves an open question. In the future, we can test the new models in other big data sets. If they perform poorly, we could use other basic models to replace the ARMA and SVM models. (2) Interval prediction results rely on point prediction, and the accuracy of interval prediction obviously decreases when extreme events occur even though

the proposed model outperforms other competing models in these periods. For the future, we could further consider other relevant factors into the proposed model to identify the uncertain information, which might increase the prediction accuracy in the period of extreme events. For example, facing the sub-prime crisis and COVID-19, one could capture and quantize the uncertainty by using some reasonable measures such as the economic policy uncertainty index and Google search volume index. These information could further identify the tendency of crude oil prices and thus help to improve forecasting ability. (3) Whether the new model is dominant in other prediction tasks of the energy field is also meaningful. For instance, wind speed forecasting and electricity load are attracted to scholars. Whereas, these time series are fundamentally different from crude oil price series because they have seasonality, which could be also an interesting future work.

CRediT authorship contribution statement

Yuan Zhao: Conceptualization, Methodology, Software, Writing – original draft, Formal analysis. **Weiguo Zhang:** Writing – review & editing, Funding acquisition, Resources. **Xue Gong:** Supervision, Writing – review & editing, Validation. **Chao Wang:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Nomenclature

ADF test	Augmented Dikey-Fuller test
VMD	variational mode decomposition
SVM	support vector machine
MAE	mean absolute error
PSO	particle swarm optimization
ARMA	autoregressive moving average mode
AMEE	average maximum envelope entropy
CEEMD	complementary ensemble empirical mode decomposition
CEEMDAN	complementary ensemble empirical mode decomposition with adaptive noise
PVMD-SVM-ARMA	the proposed model for point prediction
EMD-SVM-ARMA	hybrid model based on EMD, SVM and ARMA
CEEMD-SVM-ARMA	hybrid model based on CEEMD, SVM and ARMA
EMD-ELM-ARMA	hybrid model based on EMD, ELM and ARMA
CEEMD-ELM-ARMA	hybrid model based on CEEMD, ELM and ARMA
DM test	Diebold Mariano test
MCS	model confidence set test
MSE	mean square error
TIC	Theil inequality coefficient
EMD	empirical mode decomposition
AIC	minimum Akaike information criterion test
EEMD	ensemble empirical mode decomposition
PSO-VMD/PVMD	VMD optimized by PSO with AMEE fitness function
PVMD-SVM-ARMA-Bmax	the model corresponding to the maximum mean square error in PVMD-SVM-ARMA-B
PVMD-SVM-ARMA-B	the proposed model for interval prediction
EEMD-SVM-ARMA	hybrid model based on EEMD, SVM and ARMA

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CEEMDAN-SVM-ARMA	hybrid model based on CEEMDAN, SVM and ARMA
EEMD-ELM-ARMA	hybrid model based on EEMD, ELM and ARMA
CEEMDAN-ELM-ARMA	hybrid model based on CEEMDAN, ELM and ARMA

Appendix B. Algorithm

Algorithm 1. VMD

```

1: Initialize  $\{u_k^1\}, \{\omega_k^1\}, \lambda^1, n = 0$ 
2: repeat
3:  $n = n + 1$ 
4: for  $k = 1$  to  $K$  do
5:   Update  $\hat{u}_k$  for all  $\omega > 0$  by Eq. 3
6:   Update  $\hat{\omega}_k$  by Eq. 4
7: end for
8: Update  $\hat{\lambda}$  for all  $\omega > 0$  by Eq. 5
9: until  $\sum_{k=1}^K \frac{\|u_k^{n+1} - u_k^n\|_2^2}{\|\hat{u}_k^n\|_2^2} < \zeta$ 

```

Algorithm 2. PSO-VMD

```

1: Set parameter values of  $\omega_{\min}, \omega_{\max}, c_1, c_2, size, P_{\max}, P_{\min}, V_{\max}, V_{\min}$  and  $maxgen$ 
2: for each particle do
3:   Initialize  $P, V, pbest, Pbest, fbest, Fbest$ , that is
4:    $P = floor((P_{\max} - P_{\min}) * rand() + P_{\min})$ 
5:    $V = floor((V_{\max} - V_{\min}) * rand() + V_{\min})$ 
6:   Decompose sequence by Algorithm 1 and calculate Fitness according to Eq. 12
7:   Initialize  $pbest$  and  $Pbest$  according to  $P$ 
8:   Initialize  $fbest$  and  $Fbest$  according to Fitness
9: end for
10: for  $iter = 1$  to  $maxgen$  do
11:   Update  $\omega$  according to Eq. 13
12:   for  $i = 1$  to  $size$  do
13:     Update  $V, P, Fitness$  according to Eq. 14 and Eq. 12
14:     if  $fbest < Fitness$  then
15:        $fbest = Fitness$ 
16:        $pbest = P$ 
17:     end if
18:     if  $Fbest < fbest$  then
19:        $Fbest = fbest$ 
20:        $Pbest = pbest$ 
21:     end if
22:   end for
23: end for
24: Print  $Fbest, Pbest$ 

```

Algorithm 3. The proposed method

```

1: Input time series  $S$ 
2: %% The optimal parameters of VMD are determined according to PSO-VMD algorithm
3: Set the initial parameters of VMD
4: Sequence is initially decomposed by VMD algorithm.
5: Calculate the envelope entropy of subsequence.
6: Determine the optimal parameters of VMD by PSO algorithm with the fitness function AMEE.
7: %% Decompose sequence by the optimal parameters.
8: Decompose sequence according to the best parameters obtained by Algorithm 2, the decomposed subsequence is called  $u_i$  and the noise sequence is called  $NO$ .
9: %% Modeling
10: for  $i = 1$  to  $k$  do
11:   Subsequence  $u_i$  is made ADF test

```

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12: if ADF test of  $u_i$  show that the null hypothesis is not true then
13:   According to subsequence  $u_i$ , ARMA model is established by AIC principle, which is called Sub-Modeli.
14: else
15:   SVM model is established on the basis of subsequence  $u_i$ , which is Sub-Modeli.
16: end if
17: %% Forecasting
18: Based on rolling prediction of one-step ahead, forecasting set of subsequence  $u_i$  can be obtained by updating the input
   data of Sub-Modeli.
19: end for
20: % The prediction results of PVMD-SVM-ARMA model
21: Forecasting sets of subsequence are reconstructed to get the predicted values of original sequence.
22: % Obtain noise matrix
23: NO is resampled by Bootstrap, and noise matrix is formed by transforming the sampling results
24: % The interval prediction results of PVMD-SVM-ARMA-B model
25: Noise matrix is added into the prediction results of PVMD-SVM-ARMA model, which can get the forecasting results of
   PVMD-SVM-ARMA-B model

```

Appendix C. The details of method

C.1. The details of proposed model

The noise sequences is sampled L times repeatedly by Bootstrap, and then noise matrix of $h * L$ dimension is got, in which, h is the length of training set. According to the length m of test set, the size of noise matrix is transformed into $m * L_1$. New noise matrix NO is added into the prediction results of PVMD-SVM-ARMA model to get the interval prediction value $\tilde{x}_i$. The maximum and minimum values of PVMD-SVM-ARMA-B model are selected to make up the boundary of interval prediction. The point and interval prediction results are calculated by

$$\begin{aligned}\hat{x} &= \hat{I}_1 + \hat{I}_2 + \dots + \hat{I}_K \\ \tilde{x} &= \hat{I}_1 + \hat{I}_2 + \dots + \hat{I}_K + NO\end{aligned}\quad (C.1)$$

where $\hat{I}_i$ is the prediction sequence of i th IMF, NO is the noise information matrix, $\hat{x}$ is the predicted series of PVMD-SVM-ARMA model, and $\tilde{x}$ is the interval prediction results of PVMD-SVM-ARMA-B model. The maximum fluctuation range of interval prediction is

$$\begin{aligned}\tilde{x}_{t,l} &= \min(\tilde{x}_i) \\ \tilde{x}_{t,u} &= \max(\tilde{x}_i)\end{aligned}\quad (C.2)$$

in which, $\tilde{x}_{t,l}$ and $\tilde{x}_{t,u}$ are the lower and upper bound of interval prediction in the t th value.

The sequences corresponding to the maximum mean square error (MSE) in PVMD-SVM-ARMA-B model are taken as the comparison results. Comparing with other comparison models, it can help to verify the feasibility of proposed interval prediction. The calculation formula is

$$\tilde{x} = \max(\text{MSE}(\tilde{x}_i)), \quad (C.3)$$

where $\tilde{x}$ is called the forecasting result of PVMD-SVM-ARMA-Bmax model. $\tilde{x}_i$ is the i th sequence of matrix $\tilde{x}$, where $i = 1, 2, \dots, L_1$.

C.2. Prediction scheme

Single stage prediction is actually offline prediction, which is accomplished by rolling prediction of one step ahead. Multistage prediction is actually online real-time prediction, that is, data is updated online to get the new model and online prediction results. Assuming the training set is $x_1, x_2, \dots, x_h$ in the single stage, $x_1, x_2, \dots, x_h$ is applied to construct model and predict x_{h+1} . And then removing x_1, x_{h+2} can be gained by $x_2, x_3, \dots, x_{h+1}$. Hence the forecasting set $\hat{x}_{h+1}, \hat{x}_{h+2}, \dots, \hat{x}_{h+m}$ can be calculated in a similar fashion. In the multistage prediction, the training set is updated online, and the model is updated by Algorithm 3 to make the new prediction. The length of updating window in online real-time prediction is that of the forecasting set, namely m . Fig. C.5 provides a particular description of prediction scheme. In brief, according to the data update of multistage in Fig. C.5, S in Algorithm 3 is updated to realize online real-time prediction.

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